5

DIGITAL ELECTRONICS

Digital Electronics

27/08/12

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Sylabus

1) logical expression minimization → Boolean algebra, → K·map

2) Combinational ckts → Arithmatic ckts → HA, → HS, → FA, FS, PA, → Look Ahed Carry Adder.

(3) MUX, DMUX, Decoders, Encoders | Comparators, code converters.
> combinational

· Sequential ckts

(i) FF's, Registers, counters

(2) ADC/DAC Logic families

(3) Memory. → RAM, ROM**

> Number systems - codes - data representation. (imp)

Refrences:-

(i) Digital logic design → morish mano (unsolved questions)

(ii) ~~Digital integrated electronics~~ → Taub. & Schilling (for sequential)

(iii) Logic Design → Roth

Practice questions (20% Repetition)

GATE → ECE, EEE, EI, CS, IT → 90% PSU's Cover

IES → ECE, → EEE | IAS → 10 years paper

Boolean Algebra :- 1854 → George boolean

Logical expression minimization → Boolean Algebra → 1, 2, 3 Variables (0,1)
→ K map. 2, 3, 5 (variables) (0,1,x)
→ Quine mcclusky / Tabular method
↳ can be used for any No. of variables.

(1) A ⟶ NOT ⟶ $\overline{A}$ (or) A'

$\overline{\overline{A}} = A$

(2) AND

$0 \cdot 0 = 0$
$0 \cdot 1 = 0$
$1 \cdot 0 = 0$
$1 \cdot 1 = 1$
⟶
$$\boxed{\begin{array}{l} A \cdot A = A \\ A \cdot 0 = 0 \\ A \cdot 1 = A \\ A \cdot \overline{A} = 0 \end{array}}$$

(3) OR

$$\boxed{\begin{array}{l} A + A = A \\ A + 0 = A \\ A + 1 = 1 \\ A + \overline{A} = 1 \end{array}}$$

Ex→ SOP

Q. $AB + A\overline{B} = A(B+\overline{B}) = A$ (redundant / independent)

Q. $AB + A\overline{B}C + A\overline{B}\overline{C} = AB + A\overline{B}(1) = A(B+\overline{B}) = A$

minimum No. of NAND gate required = ?

Ans. 0

Qs POS

$(A+B)(A+C) = AC + BC$

$= A + AC + AB + BC$

$= A(1 + C + B) + BC$

$$\boxed{(A+B)(A+C) = A + BC}$$

Que $(\bar{X}$ $= \bar{X} + YZ$ 2

Que $(A+$ $)(\bar{A}+B)(\bar{A}+\bar{B})$

$\Rightarrow (A+B$ $\bar{A}+B\bar{B})$

$\Rightarrow A\bar{A} = 0$

Que $(A+B+C)(A+\bar{B}+C)(A+B+\bar{C})$

$\Rightarrow (A+B+C\bar{C})(A+\bar{B}+C)$

$= (A+B)(A$

$A+B = x$

$\Rightarrow (x+C\bar{C})(A+\bar{B}+C)$

$\Rightarrow (A+B)(A+\bar{B}+C)$

$\Rightarrow (A+B(\bar{B}+C)) = A+BC$

$$\boxed{(A+B)(A+C) = A+BC}$$

$$\boxed{\begin{matrix} A + BC = (A+B)(A+C) \\ ①\ \ ②③\ \ (①+②)(①+③) \end{matrix}}$$ $\Rightarrow$ Distribution theorem

Que $A+\bar{A}B = (A+\bar{A})(A+B) = A+B$

Que $(A+\bar{A}\bar{B}) = (A+\bar{A})(A+\bar{B}) = A+\bar{B}$

Que $\bar{A}+AB = \bar{A}+B$

Que $\bar{A}+A\bar{B} = \bar{A}+\bar{B}$

Que $AB + A\bar{B} + \bar{A}\bar{B} = A(B+\bar{B}) + \bar{A}\bar{B}$

$= A + \bar{A}\bar{B} = A+\bar{B}$

we can add any term present in expression

$\because A+A=A$

OR

Que $A\bar{B} + \bar{A}B + AB$

$\Rightarrow A(B+\bar{B}) + \bar{A}B = A+\bar{A}B = A+B$

we can add any term present in expression.

$\because \boxed{A+A=A}$

Que. $AB C + A\bar{B}C + \bar{A}BC$

$\Rightarrow$ ~~$BC + A\bar{B}C$~~ $\Rightarrow ABC + A\bar{C}$ ... $C + ABC$

$\Rightarrow$ B ... $1 + A = A$

$\Rightarrow$ ~~$BAC(A+\bar{A}) + BC(A+\bar{A})$~~ $BC(A+\bar{A}) + AC(B+\bar{B})$

$\Rightarrow AC + BC \longrightarrow$ SOP

$\Rightarrow C\cdot(A+B) \longrightarrow$ POS

Que

$AB + \bar{A}C + BC\cdot 1 = AB + \bar{A}C$ (prove)

$\Rightarrow AB + \bar{A}C + BC(A+\bar{A})$ Redundant

$\Rightarrow AB + \bar{A}C + ABC + \bar{A}BC$

$\Rightarrow AB(1+C) + \bar{A}C(1+B)$

$\Rightarrow AB + \bar{A}C$

SOP Imp

$\boxed{AB + \bar{A}C + BC = AB + \bar{A}C}$ $\Rightarrow$ Consensus Theorem ~~Redudan~~ Redundant

① Three variable

② repeat Twice

③ only one variable is Complemented / one variable uncomplimente also

AB
$\bar{A}$C
BC

Que $AB + \bar{B}C + AC = AB + \bar{B}C$

Que $AB + B\bar{C} + AC = B\bar{C} + AC$

POS Imp

$\boxed{(A+B)(\bar{A}+C)(B+C) = (A+B)(\bar{A}+C)}$

Redundant Terms

$(A+B)(B+\bar{C})(A+C) = (B+\bar{C})(A+C)$

extension $\bar{A}\bar{B} + \bar{B}\bar{C} + A\bar{C} = (\bar{A}\bar{B} + A\bar{C})$. 3

$\bar{A}B + \bar{B}\bar{C} + \bar{A}\bar{C} = \bar{A}B + A\bar{B}\bar{C}$

$(A+\bar{B})(\bar{A}+\bar{C})(\bar{B}+\bar{C}) = (A+\bar{B})(\bar{A}+\bar{C})$

Qs $(A+B)(\bar{A}+C)$ $(A+B)(A+C) = A+BC$

$\Rightarrow AC + \bar{A}B + BC$

$\Rightarrow AC + \bar{A}B$

$\Rightarrow \boxed{(A+B)(\bar{A}+C) = AC + \bar{A}B}$ ← Transposition Theorem

4 $\underset{POS}{(A+B)(\bar{A}+\bar{B})} = \underset{SOP}{A\bar{B} + \bar{A}B} \Rightarrow$ EX-OR

$\Rightarrow$ Demorgan Theorem :-

$\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$

$\overline{A+B+C} = \bar{A}\cdot\bar{B}\cdot\bar{C}$

Boolean Algebra

(1) Theorems $\rightarrow$ minimize (9) statements

(2) SOP $\rightarrow$ minimal form, $\rightarrow$ Canonical form

(3) POS

(4) Dual

(5) Complement

(6) Venn diagrams

(7) Truth tables

(8) switching circuits

SOP (sum of product) :-

ex → $ABC + \bar{A}BC + \underbrace{A\bar{B}\bar{C}}_{\text{product term}}$

min term

→ In SOP → each individual term is known as min term.

→ SOP form is used when the output of logic ckt is logic 1.

→ min term for $5 \rightarrow 101 \rightarrow A\bar{B}C$

$9 \rightarrow 1001 \rightarrow A\bar{B}\bar{C}D$

Que. For the give Truth Table minimized SOP expression is
→ output is logic 1

	A	B	Y
$\bar{A}\bar{B}$ ←	0	0	1
←	0	1	0
$A\bar{B}$ ←	1	0	1
AB ←	1	1	1

$$Y = \bar{A}\bar{B} + A\bar{B} + AB$$

$$= A + \bar{A}\bar{B}$$

$$= A + \bar{B}$$

OR $Y(A,B) = \Sigma m(0,2,3)$

Que ⇒ $f(A,B) = \Sigma m(0,1,2)$

$$\Rightarrow f = \bar{A}\bar{B} + \bar{A}B + A\bar{B}$$

$$= \bar{A} + A\bar{B}$$

$$= (\bar{A}+A)(\bar{A}+\bar{B})$$

$$= \bar{A}+\bar{B}$$

$A + \bar{A}B = A + B$ → minimal form (further cannot be minimized)

$A + \bar{A}B$

$\Rightarrow A \cdot 1 + \bar{A}B$

$\Rightarrow A(B+\bar{B}) + \bar{A}B = \boxed{A\bar{B} + \bar{A}B} \Rightarrow$ ~~Canoni~~: Canonical SOP form (each min term must contain all variables)

→ in Canonical SOP form each min term will have all the variables.

Canonical form OR Standard SOP form

Que No. of min terms present in Canonical SOP of $A + \bar{B}C$ is ...

Soln

$\cancel{A(B+\bar{B}) + \bar{B}C(A+\bar{A})}$

$\Rightarrow \cancel{(AB+A\bar{B})(C+\bar{C})} + \cancel{B}\,A\bar{B}C + \bar{A}\bar{B}C$

$\Rightarrow A(B+\bar{B})(C+\bar{C}) + \bar{B}C(A+\bar{A})$

$\Rightarrow A(BC + B\bar{C} + \bar{B}C + \bar{B}\bar{C}) + A\bar{B}C + \bar{A}\bar{B}C$

$\Rightarrow ABC + AB\bar{C} + A\bar{B}C + \cancel{B}\,A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C$

one term

$\Rightarrow$ 5 term

(Product of Sum) form :-

POS form is used ~~one~~ ~~o/p~~ when output is logic 0.

ex→ $(A+B+C)(\bar{A}+B+\bar{C})(\bar{A}+B+\bar{C})$

→ each individual term is called as max term.

ex→ $5 \rightarrow 101 \rightarrow$ max term → $(\bar{A}+B+\bar{C})$

Q. for the given Truth Table minimized POS expression is

A	B	Y
0	0	1
0	1	0 ✓
1	0	1
1	1	0 ✓

~~$Y = (\bar{A}$~~

$Y = (A+\bar{B})(\bar{A}+\bar{B})$

same

$Y = \cancel{A}\,\bar{B} + A\bar{A}$

$Y = \bar{B}$

OR

$A\bar{B} + \bar{A}\bar{B}$

$\Rightarrow \bar{B}(A+\bar{A})$

$\Rightarrow \bar{B}$

OR $Y(A,B) = \prod M(1,3)$

SOP → $Y = \bar{A}\bar{B} + A\bar{B}$

$\Rightarrow \bar{B}(A+\bar{A}) = \bar{B}$

$\Rightarrow \boxed{\sum m(0,2) = \prod M(1,3)}$

Q. $f(A,B,C) = \sum m(1,2,5,7) = \prod M(0,3,4,6)$

Note: with n variable max. possible min terms OR max terms is 2^n.

$A, B \longrightarrow$ Total No. of logic expressions $= 2^{2^n} = \underline{16}$

$\overline{A \cdot B}$	$\overline{A}B$	A	0
$A+B$	$A\overline{B}$	$\overline{A}$	1
$\overline{A}+B$	$\overline{A}\cdot\overline{B}$	B	$\overline{A}B+A\overline{B}$
$A+\overline{B}$	$\overline{A}+\overline{B}$	$\overline{B}$	$AB+\overline{A}\,\overline{B}$

$\Rightarrow$ with n variables maximum possible logic expression are $= 2^{2^n}$

if $n=2 \Rightarrow 2^{2^2} = 2^4 = 16$

$n=3 \Rightarrow 2^{2^3} = 2^8 = 256$

$n=4 \Rightarrow 2^{2^4} = 2^{16} = 2^6 \times 2^{10} = 64K$

$= 65536$

Dual

Dual expression is used to convert positive logic into Negative logic and vice versa.

logic 0 → 0V, logic 1 → +5V } +ve logic

logic 0 → −5V, logic 1 → 0V } +ve logic

logic 0 → +5V, logic 1 → 0V } −ve logic

AND gate (+ve logic)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR gate (+ve logic)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

–ve logic OR gate

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	0

= +ve logic AND gate

⇒ +ve logic AND = –ve logic OR

⇒ AND $\xleftrightarrow{\text{Dual}}$ OR

⇒ $A \cdot B \xleftrightarrow{\text{Dual}} A+B$

Dual

1) AND ⟷ OR

$\cdot \longleftrightarrow +$

2) $0 \longleftrightarrow 1$

3) Keep variables as it is.

Ex. $A\bar{B}C + AB\bar{C} + \bar{A}\bar{B}\bar{C}$

Dual ⇒ $(A+\bar{B}+C)\cdot(A+B+\bar{C})\cdot(\bar{A}+\bar{B}+\bar{C})$

Dual = $A\bar{B}C + AB\bar{C} + \bar{A}\bar{B}\bar{C}$ (original expression)

⇒ for any logical expression if two times Dual is used Result is same expression.

Self Dual

Ex $AB + BC + AC$

Dual = $(A+B)\cdot(B+C)\cdot(A+C) = (B+AC)(A+C)$

$= AB + BC + AC + AC$

$= AB + BC + AC$

⇒ in self Dual expressions if one time Dual is used Result is same expression. 6

⇒ with n variables maximum possible self dual are $2^{2^{n-1}}$.

$n = 1 \Rightarrow 2^{2^0} = 2^1 = 2$

$A \xleftrightarrow{\text{Dual}} A$

$A \cdot 1 \xleftrightarrow{\text{Dual}} (A+0)$

$\overline{A} \xleftrightarrow{\text{Dual}} \overline{A}$

AND $\xleftrightarrow{\text{Dual}}$ OR	
NAND $\xleftrightarrow{\text{Dual}}$ NOR	
NOT $\xleftrightarrow{\text{Dual}}$ NOT **	

Complement :-

$$Y = ABC + \overline{A}BC + A\overline{B}C$$

$$\overline{Y} = \overline{ABC + \overline{A}BC + A\overline{B}C}$$

$$= (\overline{ABC}) \cdot (\overline{\overline{A}BC}) \cdot (\overline{A\overline{B}C})$$

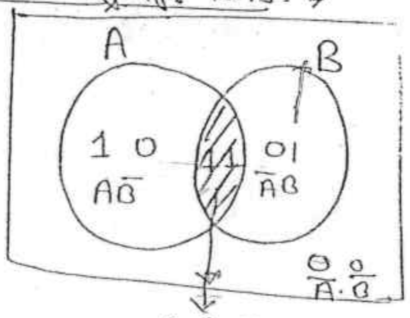
$$= (\overline{A} + \overline{B} + \overline{C}) \cdot (A + \overline{B} + \overline{C}) \cdot (\overline{A} + B + \overline{C})$$

(i) AND ⟷ OR

(ii) 1 ⟷ 0

(iii) complement each variable

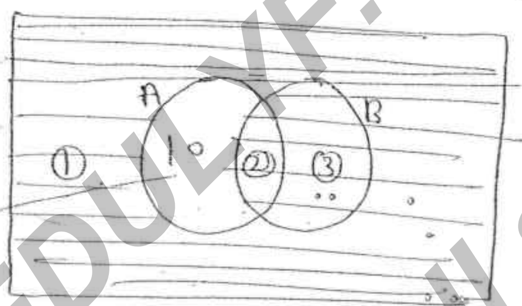
Venn diagrams: $\Rightarrow$



$A \cap B$

$A \cdot B$

Que For the given venn diagram minimize expression for shaded area shown in figure.



$\bar{A}+B$

Soln

$$f = \bar{A}\bar{B} + AB + \bar{A}B$$

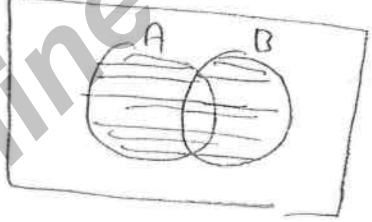
$$= \cancel{\bar{A} + \bar{A}B}$$

$$= \bar{A}\,\cancel{(B+\bar{B}) + B(A+\bar{A})}$$

$$= \cancel{\bar{A}+B}$$

$$= \bar{A} + AB = \bar{A} + B$$

Que



$A \cup B \Rightarrow A+B$ OR $f = A\bar{B} + AB + \bar{A}B$

$$= A + \bar{A}B$$

$$= A + B$$

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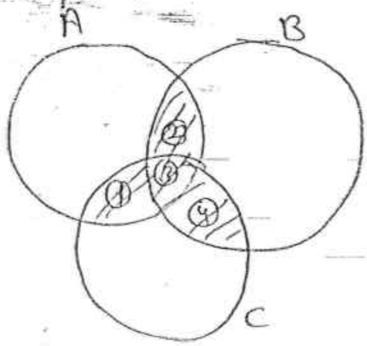
3 variable diagram :⇒



minimised expression for the venn diagram shown in fig.

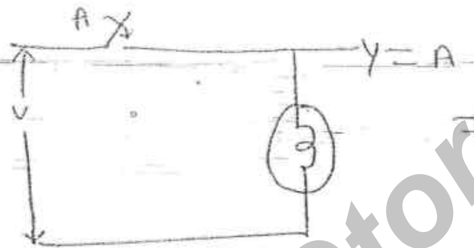
$$f = A\bar{B}C + A\bar{B}\bar{C}... $$

$$f = A\bar{B}C + \cancel{A}\ AB\bar{C} + ABC + \bar{A}BC$$

$$= A\bar{B}C + \cancel{AB + \bar{A}BC}$$

$$= AC + AB + \bar{A}BC$$

Switching Circuits :⇒

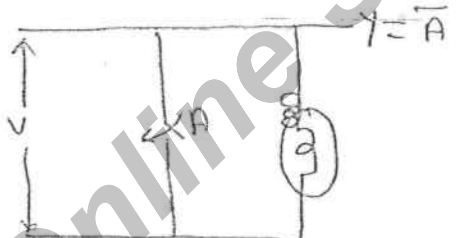


⇒

A	Y
0	→ 0
1	→ 1

⇒ $\boxed{Y = A}$

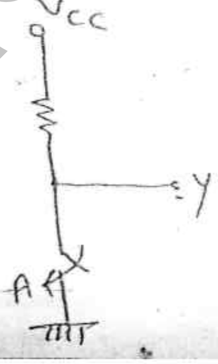
Buffer



⇒

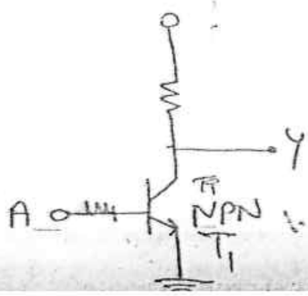
A	Y
0	→ 1
1	→ 0

⇒ $\boxed{Y = \bar{A}}$ NOT



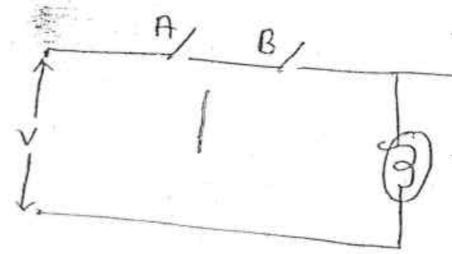
A	Y
0	1
1	0

⟹



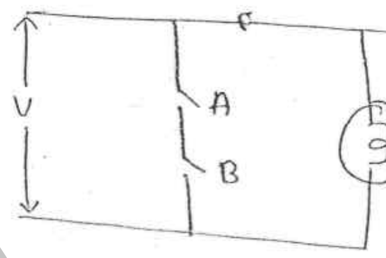
A	T_1	Y
0	off	1
1	ON	0

$Y = \bar{A}$



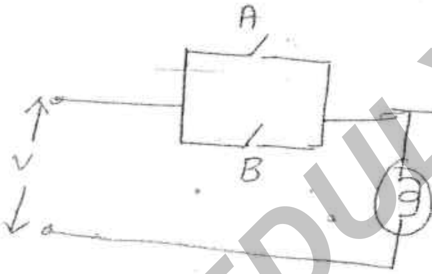
$Y = AB$
(AND)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

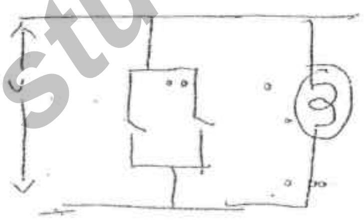


$Y = \overline{AB}$
(NAND)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

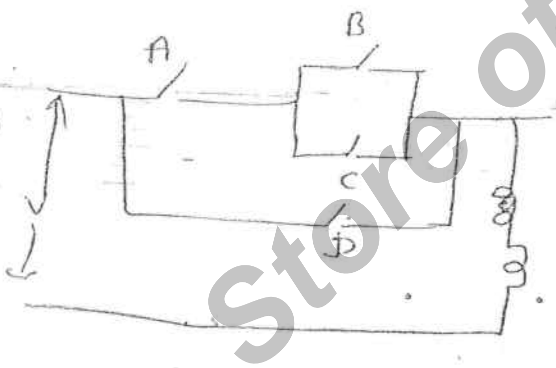


$Y = A+B$



$Y = \overline{A+B}$

Ex)



$Y = \cancel{A+}\ A\cdot(B+C)+D$

Three paths for bulb to be on

Ex)

$Y = A(B+C)\cdot\overline{D}$

Statements :⇒

Qs The a logic circuit have three inputs A, B, C and output is Y. Then Y is logic 1 for the following combination

1. A and C are true
2. B and C are falls
3. A, B and C are true
4. A, B and C are falls

Sol^n

$$Y = AC + \bar{B}\bar{C} + ABC + \bar{A}\bar{B}\bar{C}$$

$$= AC + \bar{B}\bar{C}$$

GATE & IES

Qs a logic ckt have three input A, B, C and output f, f is logic 1 when majority No. of inputs are at logic 1. then minimized exp. for f is

Sol^n

A	B	C	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

min terms $f = \Sigma m(3, 5, 6, 7)$

$\Sigma(3,5)$

$$f = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$= BC + AC + AB$$

Qs if minority functions → $\overline{AB} + \overline{BC} + \overline{AC}$

Logic gates :⇒

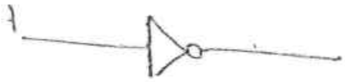
Basic Building Blocks:-

$\underbrace{\text{NOT, AND, OR}}_{\text{basic gates}}$, $\underbrace{\text{NAND, NOR}}_{\text{universal gate}}$, $\underbrace{\text{EX OR, EXNOR}}_{\text{Arithmatic ckt}}$

$\underbrace{\text{Comparator}}$

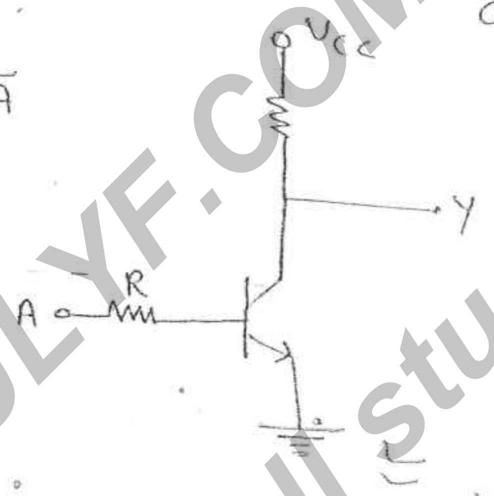
parity generator/checker
Code convertor

NOT gate



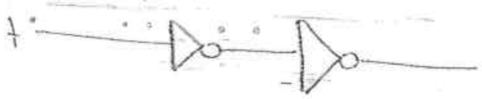
Y = $\bar{A}$

A	Y
0	1
1	0

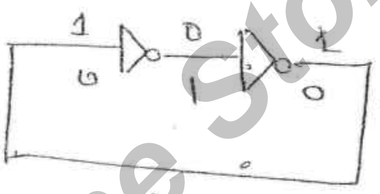


A	T_1	Y
0	off	1
1	ON	0

Ex



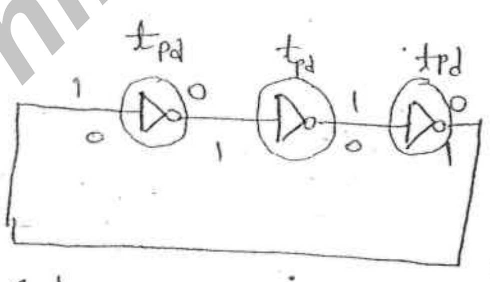
Y = A (Buffer)

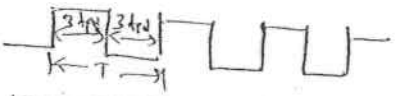


→ Stable state
→ Stable state

⇒ Two stable state
⇒ Bi-stable multivibrator
OR Basic memory ckt

Ex





$f = \frac{1}{T}$

$T = 6\, t_{pd}$
$T = 2 \times 3 \times t_{pd}$

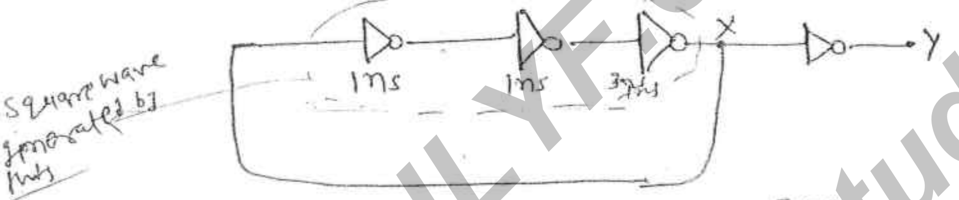
⇒ Astable multivibrator
OR ⇒ Square wave generator
OR ⇒ Clock generator
OR ⇒ Ring oscillator

in Ring oscillator ~~Timp~~ Time period of generated square wave is given by

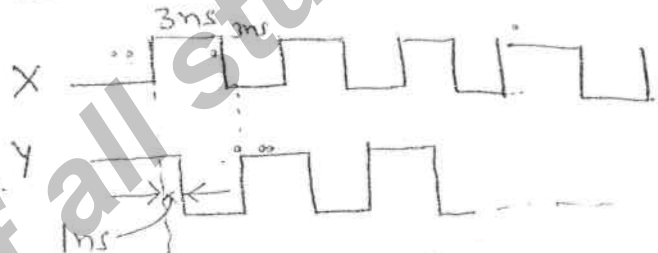
$$\boxed{T = 2N t_{pd}}$$

where $N \rightarrow$ No. of inverters connected in feedback.

⇒ **Que** In the ckt shown in fig. $t_{pd} = 1ns$ then Time period of generated square wave



$T = 6ns$



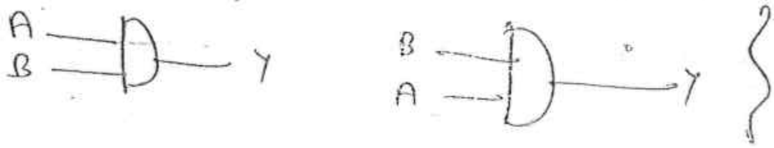
AND Gate :⇒

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

→ An AND gate follows both commutative and Associative laws.

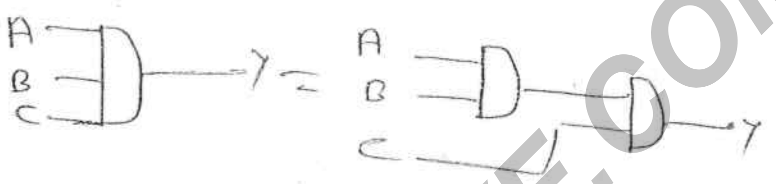
Commutative/ a... ...ve

$AB = BA$



Commutative

$ABC = (AB)C = (AC)B = (BC)A$

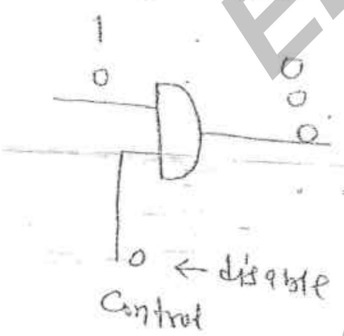


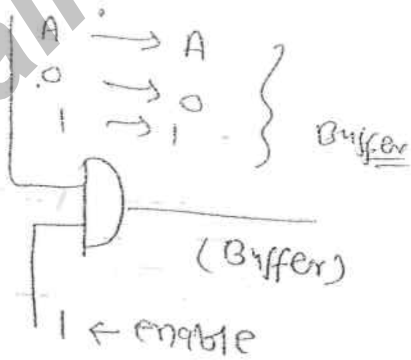
Associative

⇒ 

series switch

Enable/ disable



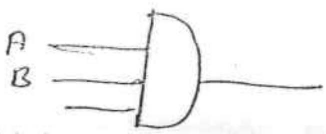


Strobe 0 → disable, strobe 1 → enable } for AND gate

→ Strobe is used to check the behaviour of the logic gate.

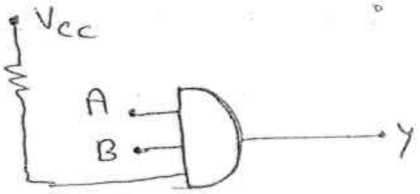
→ In TTL logic family ~~if~~ any input is open or floating it will access logic 1.

In ECL logic family open or floating input will Access logic 0.



→ In an AND gate unused input can be

(i) connected to logic '1' OR pull up. (best)



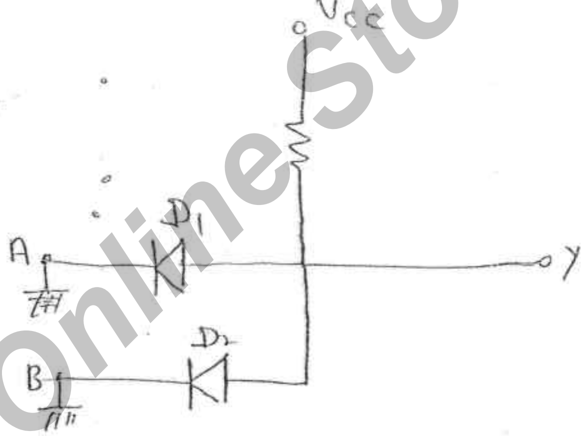
(ii) Connected to any one of used input

(fanout will reduce)

(iii) if it is TTL then unused input can be open or float. .. (N.M↓)

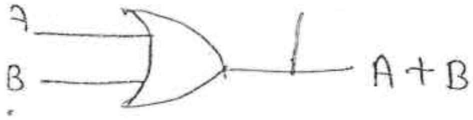
Among these the best way is to connect the logic '1' or pull up.

Que AND GATE using Diode :-

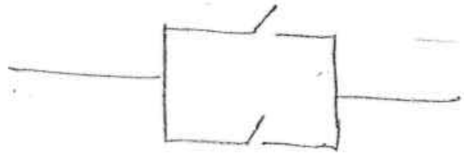


A	B	D_1	D_2	Y
0	0	ON	ON	0
0	1	ON	off	0
1	0	off	ON	0
1	1	off	off	1

OR Gate :-



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

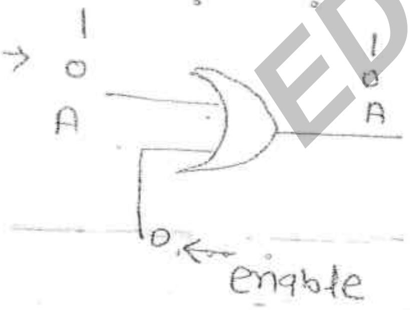
→ 

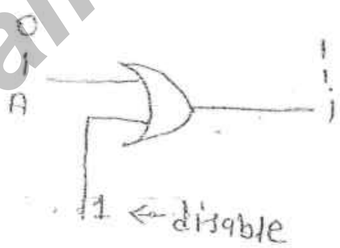
parallel switch

→ $A+B = B+A \longrightarrow$ Commutative law

$A+B+C = (A+B)+C \longrightarrow$ Associative law

OR gate follows both Commutative and Associative law.

→ 



$\Rightarrow$ OR → e → 0
d → 1

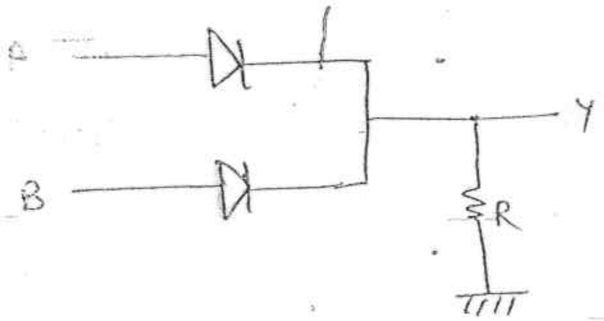
→ In OR gate unused input can be connected to

(i) Connected to logic '0' (or) pull down

(2) Can be connected to one of used input

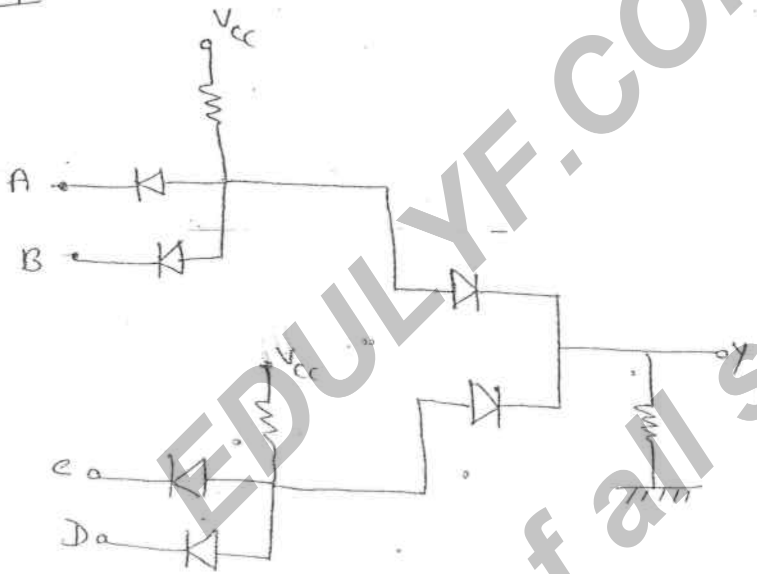
(3) if it is ECL, then unused i/p can be open/float.

OR Gate using Diode



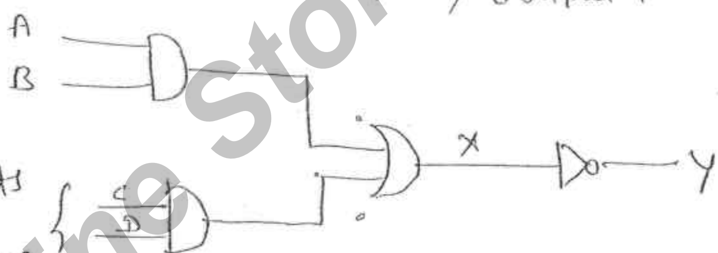
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

IES 07
Que



$Y = AB + CD$ Ans

Que
(a) In the ckt shown in fig. is TTL AND-OR-inverter for the given ~~output~~ inputs, output is —



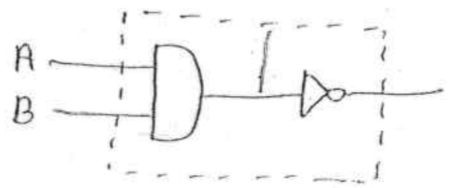
Inputs are floating {

If $C = 1, D = 1$ $\Rightarrow Y = 0$.

(b) For ECL, $C = 0, D = 0$

$\Rightarrow x = AB, \Rightarrow Y = \overline{AB}$

NAND GATE :→

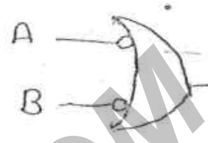


=

$\overline{AB} = \overline{A} + \overline{B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

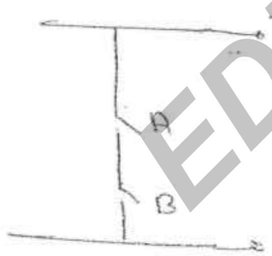
=

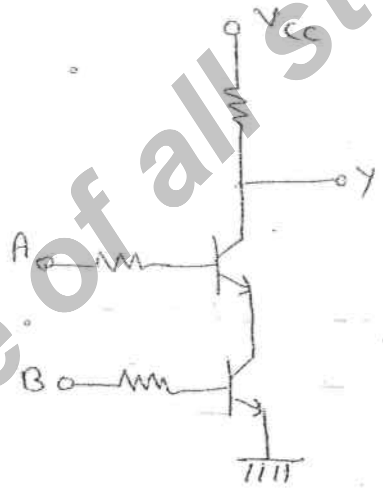


$\overline{A} + \overline{B}$

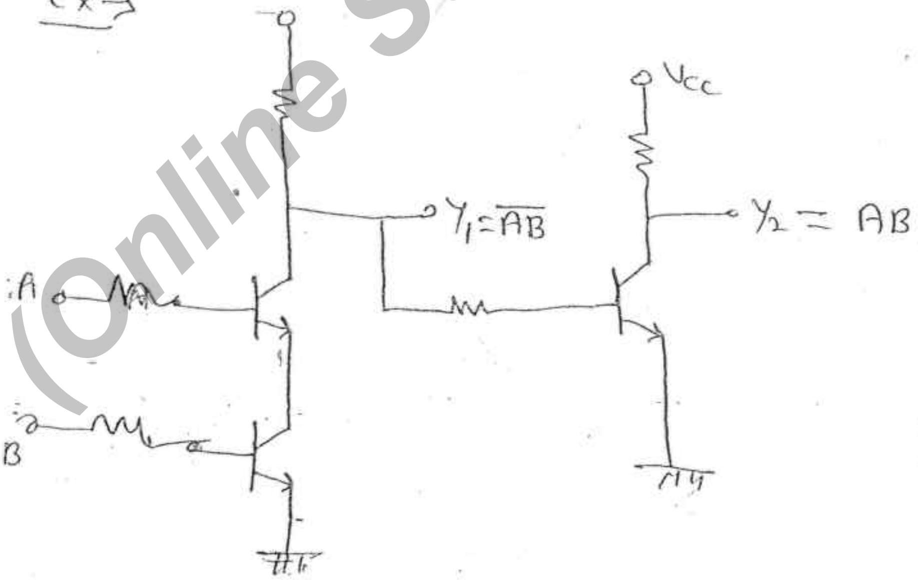
Bubbled OR

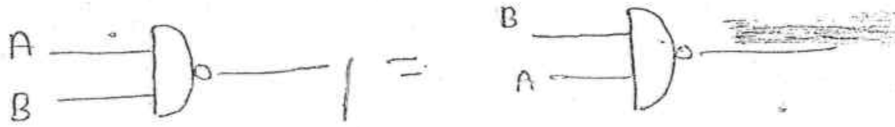
→ ~~Switch~~ NAND gate equivalent switch ckt and transistor ckt



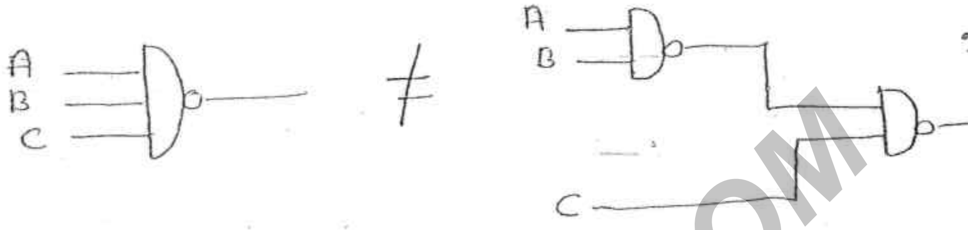


Ex →





Commutative



$\overline{(\overline{AB})C} = AB + \overline{C}$

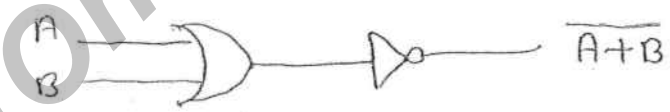
⇒ NAND gate follows Commutative law but did not follow associative law.

→

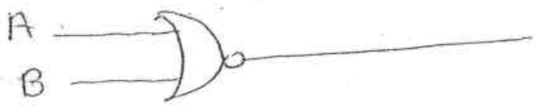
⇒ AND, NAND } e → 1, d → 0

⇒ unused input in NAND gate can be connected similar to unused input in AND gate.

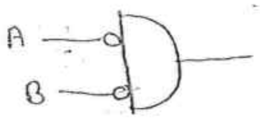
NOR Gate :⇒



$\overline{A+B}$



$\overline{A} \cdot \overline{B}$



$\overline{A} \cdot \overline{B}$

Bubbled AND gate

A	B	Y
0	0	1
0	1	0
1	0	0

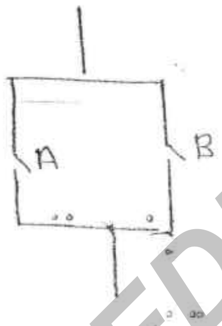
$$\left.\begin{matrix}\text{OR}\\ \text{NOR}\end{matrix}\right\} \begin{matrix} e \rightarrow 0 \\ d \rightarrow 1 \end{matrix}$$

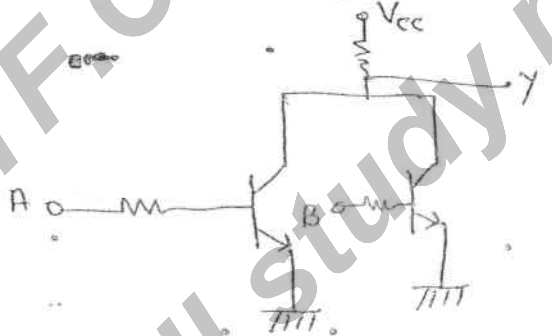
$\overline{A+B} = \overline{B+A}$ Commutative

$\overline{A+B+C} \neq \overline{\overline{A+B}+C}$

=> NOR Gate follows Commutative law but Not follows the Associative law.

=>





A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

=> Unused input in NOR gate can be connected similar to OR gate.

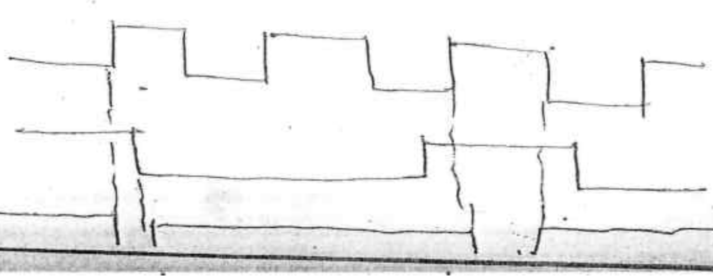
(30) $A[B+C(\overline{AB}\cdot\overline{AC})] = AB + AC(\overline{AB}\cdot\overline{AC})$ $= 0$

$A[B+C((\bar{A}+\bar{B})\cdot(\bar{A}+\bar{C}))]$

$AB+AC(\bar{A}+\bar{B}\bar{C})$

$AB+0+0 = \underline{AB}$

(31)

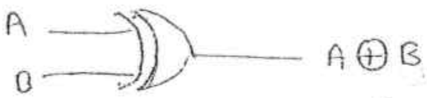


5 Y = $\left[\quad = \frac{t_{ON}}{t_{ON}+t_{off}} \right]$ 13

(8) (1) $t_{ON} = 1.7 - 0.4$, $t_{off} = 2.2 - 1.7 = 0.5$

29/08/2011

Ex-OR → Exclusive OR gate → XOR :-



$= \bar{A}B + A\bar{B}$

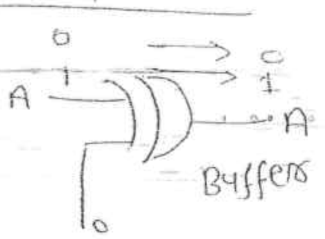
$(A+B)\cdot(\bar{A}+\bar{B})$

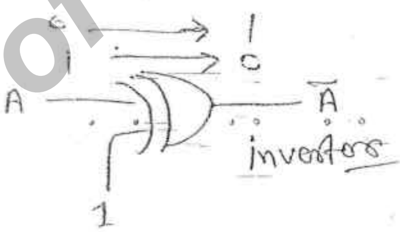
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

⇒ if $A = B \Rightarrow Y = 0$

$A \neq B \Rightarrow Y = 1$

Enable/disable





$A \oplus A = 0$

$A \oplus \bar{A} = 1$

$A \oplus 0 = A$

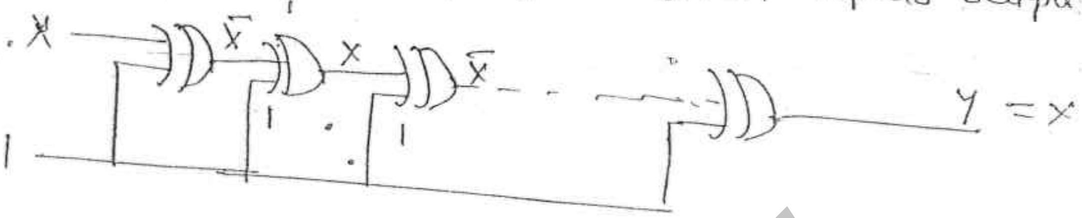
$A \oplus 1 = \bar{A}$

Que. if $A \oplus B = C$ then $A \oplus C = B$

& $B \oplus C = A$

$A \oplus B \oplus C = C \oplus C = 0$

2. The ckt shown in fig. have cascade Ex-OR gates then for given inputs output Y = ?

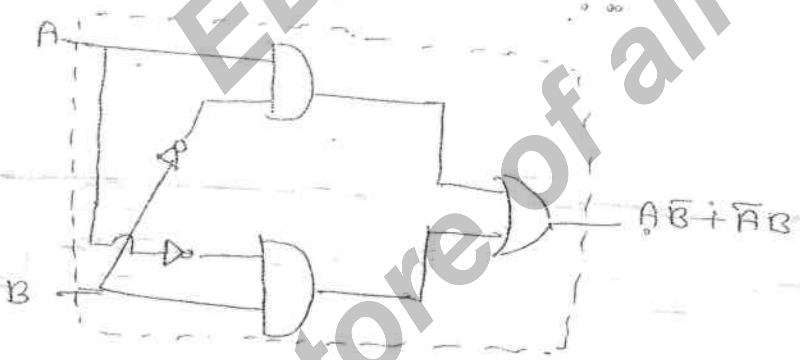


if $A \oplus A = 0$ then $A \oplus A \oplus A = A$

and $A \oplus A \oplus A \oplus A = 0$

$A \oplus A \oplus A \oplus \cdots\cdots n = A$ n is odd
$= 0$ n is even

$Y = A \oplus B = \bar{A}B + A\bar{B}$



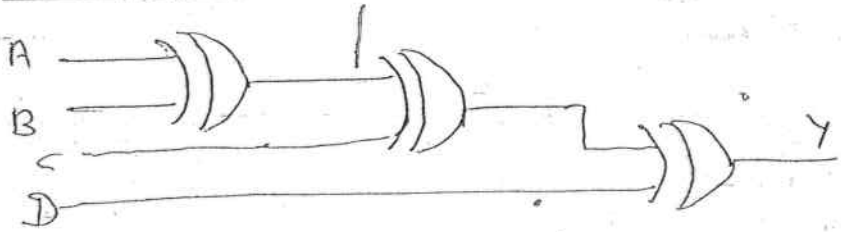
⇒ Ex-OR gate follows both Commutative and associative laws.

⇒ EX-OR gate is available with only 2 inputs, in IC format.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

in Ex-OR gate output is one when No. of one's at the input is odd Number. (It is known as odd No. of one's detector).

parity generator



4 bit parity generator

$\Rightarrow A \oplus B \oplus C$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

→ out is '1' when No. of 1's is odd

$$\boxed{Y = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC}$$ → cannot minimized

$= \cancel{\bar{A}C + \bar{B}C}$

001 010 100 111

$= \sum m(1,2,4,7)$

Que Let A = 1011, B = 1101 is two four bit data if following operations are performed in sequence.

$A \Leftarrow A \oplus B$

$B \Leftarrow A \oplus B$

$A \Leftarrow A \oplus B$

then A, B respectively

(a) A = 1011 B = 1101 (b) A = 0000 ; B = 0000

(c) A = 1101 ; B = 1011 (d) A = 1111 B = 1111

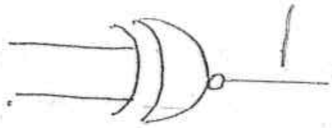
Soln

$B = A \oplus B \oplus B \Leftarrow A$

$A \Leftarrow A \oplus B \oplus A \Leftarrow B$

data interchange

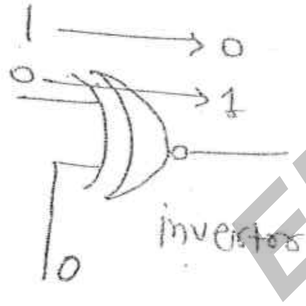
Ex-NOR Gate :→



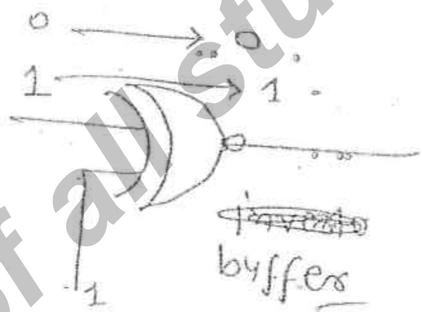
$$Y = A \odot B$$
$$= \bar{A}\bar{B} + AB$$
$$= \cancel{(\bar{A}+\bar{B})(\bar{A}}$$
$$= (A+\bar{B})(\bar{A}+B)$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

$\Rightarrow A = B \Rightarrow 1$ → Coindcedence logic gate
→ Equivalence gate



$A \neq B \Rightarrow Y = 0$



$\Rightarrow A \odot A = 1$

$A \odot \bar{A} = 0$

$A \odot 0 = \bar{A}$

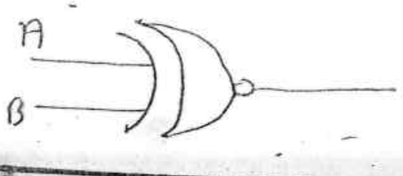
$A \odot 1 = A$

$\bar{A} \oplus \bar{B} = A \odot B$

$\Rightarrow \boxed{A \oplus \bar{B} = A \odot B}$ $\boxed{\bar{A} \oplus B = A \odot B}$

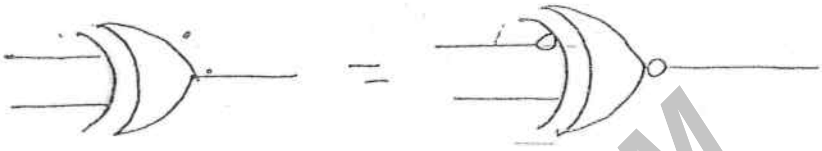
$\Rightarrow (\bar{x}y + \bar{y}\bar{x})$

$\Rightarrow (\bar{x}\bar{y} + \bar{\bar{y}}\bar{\bar{x}}) =$

$\Rightarrow$  $=$ 

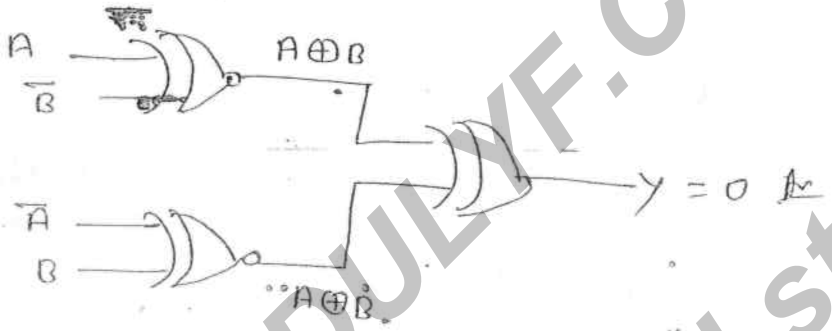
Similarly

$$\overline{A \odot B} = A \oplus B$$
$$A \odot \overline{B} = A \oplus B$$

⇒ 

$$\boxed{\overline{A} \oplus \overline{B} = A \oplus B}$$

Q. $Y = ?$



Q.

$$A \oplus B \oplus 1 = A \oplus \overline{B} = A \odot B$$

Q. $\boxed{A \oplus B \oplus AB = A + B}$

Sol ⇒ $(A \oplus B) \oplus AB = (A\overline{B} + \overline{A}B) \oplus AB = (A\overline{B} \oplus AB) + (\overline{A}B \oplus AB)$

⇒ $A\overline{B}\,\overline{AB} + \overline{A\overline{B}}AB + \overline{A}B\,\overline{AB} + \overline{\overline{A}B}AB = A\overline{B}(\overline{A}+\overline{B}) + (\overline{A}+B)AB + \overline{A}B(\overline{A}+\overline{B}) + (A+\overline{B})AB$

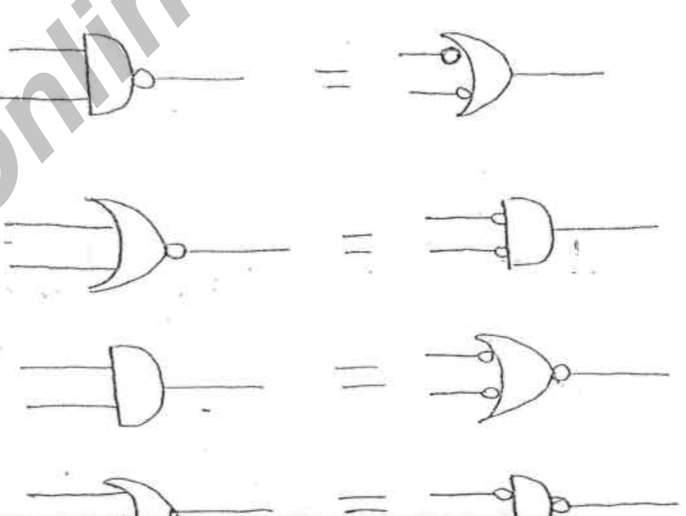
⇒ $A\overline{B} + AB + \overline{A}B + AB \Rightarrow A(B+\overline{B}) + B(A+\overline{A}) = A+B$

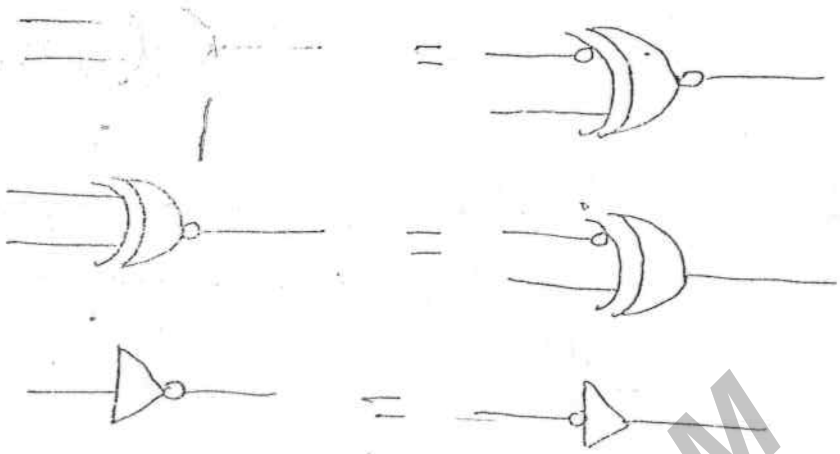
$\overline{A}B + A\overline{B} + AB$

A \ B	0	1
0	0	1
1	1	1

$\boxed{Y = A+B}$

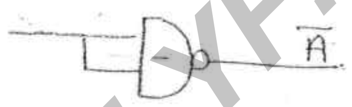
⇒ alternative symbols





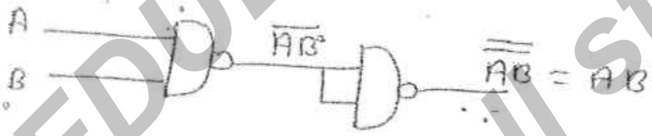
⇒ Proof: NANA as Universal Gate :

NOT →



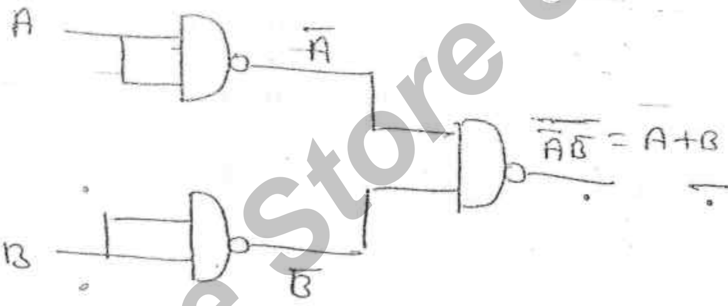
—— ① No. of NAND gate required to invert

AND



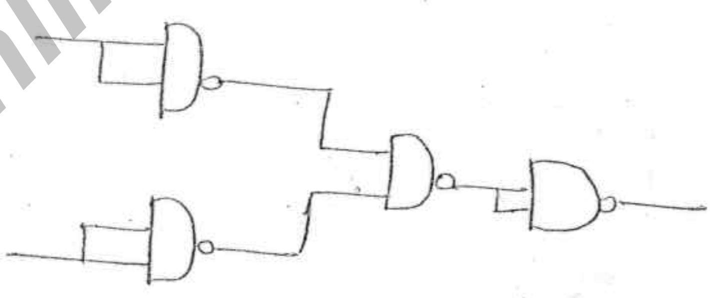
→ 2 NAND gate

OR



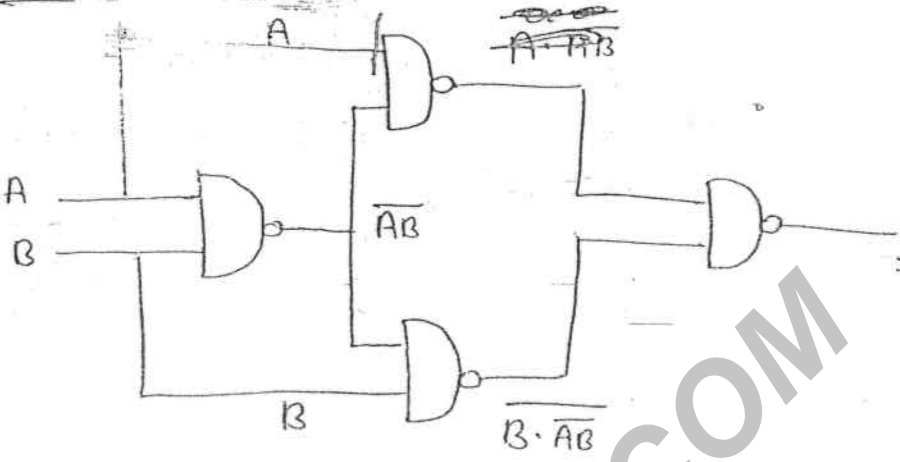
→ 3 NAND gate

NOR



→ 4 NAND gate

E



$\overline{(\overline{A\cdot\overline{AB}})\cdot(\overline{B\cdot\overline{AB}})}$

$= A(\overline{AB}) + B(\overline{AB})$

$= A(\overline{A}+\overline{B}) + B(\overline{A}+\overline{B})$

$= A\overline{B} + \overline{A}B$

$\rightarrow$ 4 ~~5~~ NAND gate

X-NOR $\rightarrow$ 5 NAND gate

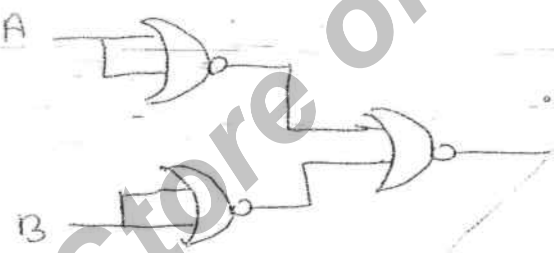
$\Rightarrow$ NOR as Universal gate

NOT $\Rightarrow$

$\overline{A+A} = \overline{A} \longrightarrow$ 1 NOR gate

AND $\Rightarrow$

=

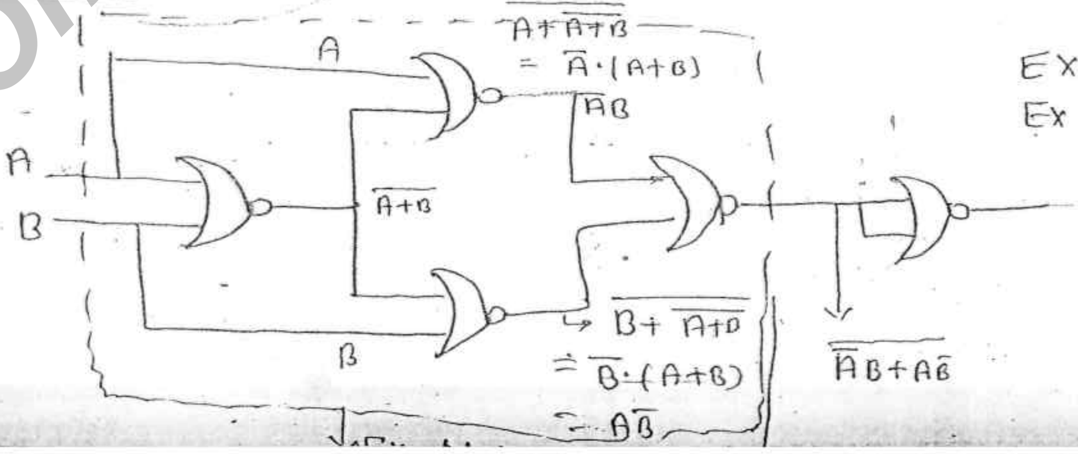


$\rightarrow$ 3 ~~NAND~~ NOR gate

OR

$\overline{\overline{A+B}} = A+B$

Ex-OR



EX-OR $\rightarrow$ 5 NOR.

EX-NOR $\rightarrow$ 4 NOR g.

⇒

Logic gate	No. of NAND	No. of NOR
NOT	1	1
AND	2	3
OR	3	2
EX-OR	4	5
EX-NOR	5	4

Note ⟶

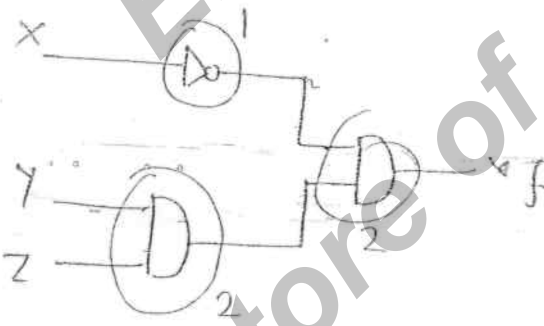
NOR ⟶ 4 NAND gates

NAND ⟶ 4 NOR

Que To implement $\bar{x}yz$ min. No. of two input NAND gate is required when only x, y, z inputs are available.

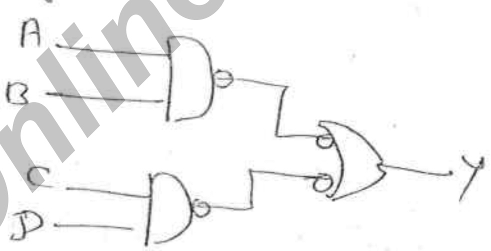
Soln (a) 3 (b) 4 (c) 5 (d) 7

Sol



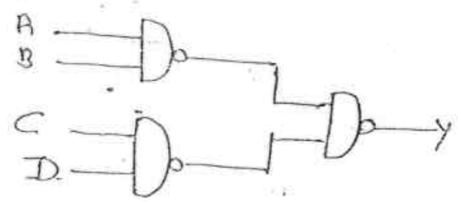
⇒ 5 NAND gate

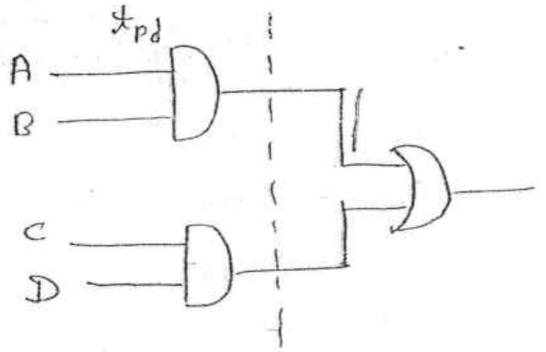
Que To implement $AB + CD$ min. No. of 2-input NAND gate required.



⇒ 3 NAND gate

⇒





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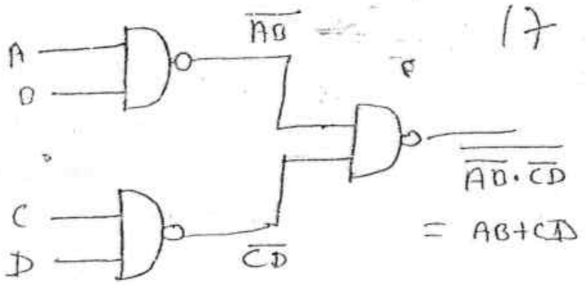
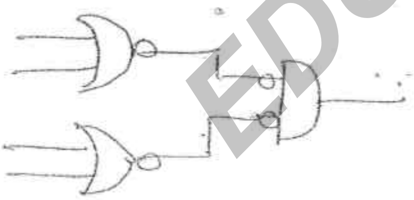


Two-level AND-OR = Two-level NAND-NAND

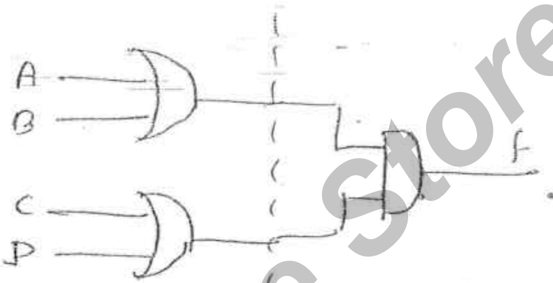
⇒ SOP expression can be implemented with minimum No. of NAND gates alone.

<u>Que</u> $(A+B)(C+D) \longrightarrow$ 3 NOR
(POS)



= ⇒ 3 NOR

SOP → NAND
POS → NOR
better to use



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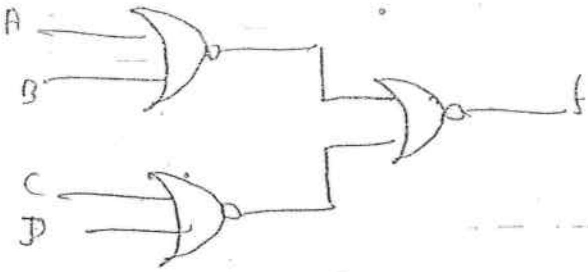


Two level OR-AND = two level NOR-NOR

<u>Que</u> $(\bar{x}+\bar{y})(w+z)$ How to implement this min. No. of
H.W two input NAND gates Required. ~~Ans~~ $\boxed{\text{Ans \# 4}}$
4 NAND gate

Number systems → Codes → data representation

(i) Conversion
(ii) Arithmetic operation
(iii) Complements

BCD
Excess 3 code
Gray code
2421

+ve -ve
Sign magnitude
1's compl
2's

Number system - code

weighted → 2 5 2 ; 100 10 1 ; positional weightage
ex → Binary, octal, Decimal, BCD
1:1 → (face, and place value)

unweighted → No positional weightage
ex → gray code, Excess code
only (face value)

⇒ $(101)_r$ → base (OR) radix

r → r different digits → 0 ... r-1

→ a No. system with base r will have r different digits and they are from 0 to r-1.

Number system	Different Digit
Binary	0,1
octal	0,1, --- 7
Decimal	0,1, ---- 9
16	0, --- 9, A, B, C, D, E, F
4	0, 1, 2, 3
12	0, --- 9, A, B

Conversion

Decimal to other

→ To convert Decimal No. into any other base or r divide integer part and multiply fractional part with base r.

ex→ $(25.625)_{10} \longrightarrow (\quad)_2$

2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

$0.625 \times 2 = 1.25 \longrightarrow 1$

$0.25 \times 2 = 0.5 \longrightarrow 0$

$0.5 \times 2 = 1.0 \longrightarrow 1$

$\Rightarrow (25.625)_{10} = (11001.101)_2$

ex→ $(17.25)_{10} = (\quad)_2$

2	17	
2	8	1
2	4	0
2	2	0
2	1	0
	0	1

$0.25 \times 2 = 0.5 \longrightarrow 0$

$0.5 \times 2 = 1.0 \longrightarrow 1$

$\Rightarrow (17.25)_{10} = (10001.01)_2$

ex→ $(25.625)_{10} = (\quad)_8$

Sol^n

8	25	
8	3	1
	0	3

$0.625 \times 8 = 5.0$

$\Rightarrow (25.625)_{10} = (31.5)_8$

$(25)_{10} = (31)_8$

$(25.625)_{10} = (\quad)_{16}$

16	25	
16	1	9
	0	1

$0.625 \times 16 = \cancel{10.000}\ \overset{A}{10.000}$

$\Rightarrow (25.625)_{10} = (19.A)_{16}$

ex → $(254)_{10} \rightarrow (\quad)_{16}$

16	254	
16	1	4
	0	14 → E

$\Rightarrow (254)_{10} = (\cancel{E4})_{16}$
$= (FE)$

16	254	
16	15	→ E (14)
	0	→ F

ex → $(25)_{10} = (\quad)_4$

4	25	
4	6	1
4	1	2
	0	1

$= (121)_4$

⇒ others to Decimal : ⇒

$(x_2 x_1 x_0 . x_{-1} x_{-2})_r = (\quad)_{10}$

$x_2 \to r^2,\ x_1 \to r^1,\ x_0 \to r^0,\ x_{-1} \to r^{-1},\ x_{-2} \to r^{-2}$

To convert any other base r into Decimal multiply each digit with positional weightage than add.

$$(\quad)_r = (\quad)_{10}$$

$$= x_2 r^2 + x_1 r^1 + x_0 r^0 + x_{-1} r^{-1} + x_{-2} r^{-2} + \cdots$$

$$= \sum_{i=-n}^{m} x_i r^i$$

ex → $(11010.11)_2 \longrightarrow (\ )_{10}$

$\Rightarrow 2^4 + 2^3 + 0 + 2^1 \times 1 + 2^0 \times 0 + 1 \times 2^{-1} + 1 \times 2^{-2}$

$\Rightarrow 16 + 8 + 1 = 25 \qquad \frac{1}{2} + \frac{1}{4} = \frac{2+1}{4} = \frac{3}{4} = 0.75$

$= (25.75)_{10}$

ex → $(37.4)_8 \longrightarrow (\ )_{10}$

$\Rightarrow 3 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1}$

$\Rightarrow 24 + 7 + 0.5 \Rightarrow (31.5)_{10}$

ex → $(47.4)_{16} = (\ )$

$\Rightarrow 64 + \cancel{7 \times 16} + \frac{1}{4} \Rightarrow \cancel{80}\ (71.25)_{10}$

ex $(BAD)_{16} \longrightarrow (\ )_{10}$

$\underset{11}{B}\ \underset{10}{A}\ \underset{13}{D}$

$\Rightarrow 11 \times 16^2 + 10 \times 16 + 13$

$\Rightarrow 11 \times 256 + 160 + 13$

$\Rightarrow \cancel{\ \ } + 160 + 13 =$
2816

$= (2989)$

ex → $(CAD)_{16} \longrightarrow (\ )_{10}$

$(DAD)_{16} \longrightarrow (\ )_{10}$

$(FACE)_{16} \longrightarrow$ //

$(CAFE)_{16} \longrightarrow$ //

ex→ $(23)_4 \longrightarrow (11)_{10}$

$2\times4^1 + 3\times4^0$

$\Rightarrow 8+3=11$

⇒ octal → binary

to convert octal No. into binary each digit is represented with 3-bit binary format.

octal	→	binary
0	→	000
1	→	001
⋮		
7	→	111

ex→ $(2|7|4)_8 = (\ )_2$

$= (010\ 111\ 100)_2$

~~ex→~~ Binary to octal

$(10110110.11)_2 \longrightarrow (\ )_8$

$\Rightarrow (266.6)_8$

Hex to binary

each digit is represented with 4 bit binary format.

ex→ $(CAD)_{16} = (\ )_2$

C→

A→10
B→11
C→12
D→13
E→14
F→15

$\Rightarrow (1100\ 1010\ 1101)_2$

No. of 1's → 7

⇒ Hex → octal

binary

20

ex→ $(FACE)_{16} = (\ )_8$

↳ $(1111\ 1010\ 1100\ 1110)_2$

$= (\cancel{175276})_8 \quad (175316)_8$

⇒ **Arithmetic Operation** :⇒

Binary → add, sub, mul

octal → add, SUB

Hex → add, sub

$(\ )_r$ → add, sub

Binary

1+0 = 0

0+1 = 1

1+1 = 10

1+1+1 = 11

ex→ sum of two binary No.'s 10110 and 11011

$$\begin{array}{r} 111 \\ 10110 \\ 11011 \\ \hline \underbrace{1}_{\text{Carry}}\ \underbrace{10001}_{\text{sum}} \end{array}$$

ex→ substract two binary No.'s 11011 and 10110

Soln

$$\begin{array}{r} 11011 \\ -\ 10110 \\ \hline 00101 \end{array}$$

multiplication

$$\begin{array}{r} 1011 \times 101 \\ \hline 1011 \\ 0000 \\ 1011 \\ \hline 110111 \end{array}$$

ex → $(1111)_2 \times (1111)_2$

⇒

$$\begin{array}{r} 1111 \times 1111 \\ \hline 1111 \\ 1111 \\ 1111 \\ 1111 \\ \hline 11100001 \end{array}$$

4 → 1's → 100

6 → 110

(22) $(10w1z)_2 \times (15)_{10} = (y01011001)_2$

⇒

$$\begin{array}{r} 10w1z \times 1111 \\ \hline 10w1z \\ 10w1zx \\ 10w1zxx \\ 10w1zxxx \\ \hline y \ldots 001 \end{array}$$

Ans $x = y = z = 1$

Octal (0 — 7)

(1) Addition

1+1 = 2

1+2 = 3

1+6 =

1+6 = 7

1+7 = 10

$(8)_{10} \to (10)_8$

8 | 8
8 | 1 | 0
  | 0 | 1

→ $(10)_8$

$7+7 = (14)_{10} = (16)_8$ 4

8	14	
8	1	6
	0	1

ex→ sum of two octal numbers 567 and 243

$$\begin{array}{r} 5\,6\,7 \\ +\,2\,4\,3 \\ \hline 1\,0\,3\,2 \end{array}$$

$7+3 = (10)_{10} = (12)_8$

8	10	
8	1	2
	0	1

$(11)_{10} \rightarrow (13)_8$

ex→ Substract 754 and 243

$$\Rightarrow \begin{array}{r} 7\,5\,4 \\ -\,2\,4\,3 \\ \hline 5\,1\,1 \end{array}$$

ex,

$$\begin{array}{r} (7\,2\,4)_8 \\ (-\,5\,6\,3)_8 \\ \hline 1\,4\,1 \end{array}$$

Hexadecimal :⇒

$1+1=2$
$1+2=3$
→
$1+9=A$
$1+A=B$
$1+B=C$
$A+A=14$ Not(E)
$A+B=15$

A + A
$10+10 = (20)_{10} = (14)_{16}$

16	20	
16	1	4
	0	1

ex, :

$$\begin{array}{r} 2\;6\;8\;9 \\ +\,5\;4\;3\;7 \\ \hline 7\;A\;C\;0 \end{array}$$

16	16	
	1	0
		1

$(16)_{10} = 10$

ex→ Substract two Hexdecm No.

$$\begin{array}{r} 9\ 5\ 4\ 3 \\ 2\ 6\ 7\ 4 \\ \hline 6\ E\ C\ F \\ \hline \end{array}$$

(8)

Complements :-

r < (r−1)'s complement / r's Complement

Binary < 1's comp / 2's   Octal < 7's c / 8's   decimal < 9's / 10's

Hexadecimal < F's → (r−1)'s complement / 16's → r's Compleme..

(r−1)'s complement :⇒

1's Comp → 1101 ⇒

$$\begin{array}{r} 1111 \\ -1101 \\ \hline 0010 \\ \hline \end{array}$$

→ To determine (r−1)'s complement Ist write max. No. Number possible in the given in the base then Substract the Number.

(i) ~~1's comp~~ 7's complem of 247 is

$$\begin{array}{r} 777 \\ -247 \\ \hline 530 \\ \hline \end{array}$$

9's complement of 865 is

$$\begin{array}{r} 999 \\ -865 \\ \hline 134 \\ \hline \end{array}$$

⇒ 1's complement of 2689 is

$$\begin{array}{r} FFFF \\ -2689 \\ \hline D976 \end{array}$$

⇒ r's complement :

To determine r's complement I^{st} write $(r-1)'s$ complement then add 1 to LSD.

(Right most place) → Least significant Digit. $\boxed{BIT \rightarrow \text{binary digit}}$

ex → Determine 2's complement of binary No. 101011

→ 1's comp. 010100

$$\begin{array}{r} 010100 \\ 1 \\ \hline 010101 \end{array}$$

ex → determine 2's Comp. of 1101.11

Soln ⇒ 1's Comp. = 0010.00

2's Comp = 0010.00

$$\begin{array}{r} 0010.00 \\ 1 \\ \hline 0010.01 \end{array}$$

direct method *

0010.01

8's

ex ⇒ determine 8's Complement 2670

Soln 7's com

$$\begin{array}{r} 7777 \\ 2670 \\ \hline 5107 \end{array}$$

8's comp. =

$$\begin{array}{r} 5107 \\ 1 \\ \hline 5110 \end{array}$$

10's

ex → 10's comp. of 5690

9's →

$$\begin{array}{r} 9999 \\ -5690 \\ \hline 4309 \end{array}$$

10's →

$$\begin{array}{r} 4309 \\ +1 \\ \hline 4310 \end{array}$$

16's complement of 5689

	F F F F
−	5 6 8 9
	A 9 7 6

16's Comp ⇒ A 9 7 6
+ 1
A 9 7 7

Digital Circuits

Combinational

⇒ present output depends only on present input

⇒ No feed back

⇒ No Memory

ex → HA, FA, MUX, Decoder

Sequential

⇒ present o/p → present i/p, → Previous/past o/ps

⇒ feed back

⇒ Memory

ex → ff / Regi / counter.

Combinational Ckt Design :⇒

(1) Identify inputs and outputs of logic circuit.
2) Construct truth table
3) write logical operation in SOP (or) POS form
4) minimize logical expressions.
5) Implement logic ckts.

Arithmetic ckt

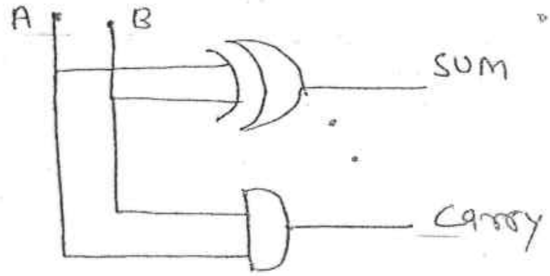
(i) Half Adder

Truth table

A	B	sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

(3). ⇒ $SUM = \bar{A}B + A\bar{B} = A \oplus B$

$Carry = AB$



HA

(i) logical exp. for sum → $A \oplus B$

(2) logical exp. for carry → AB

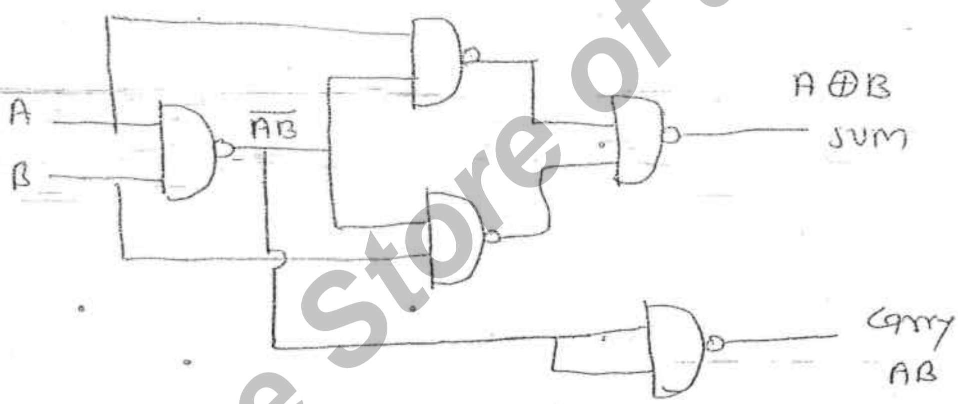
(3) minimum No. of NAND gate → 5

(4) " " " NOR gate → 5

(5) No of mux

(6) No of Decoder.

⇒ Half Adder using ~~NAR~~ NAND

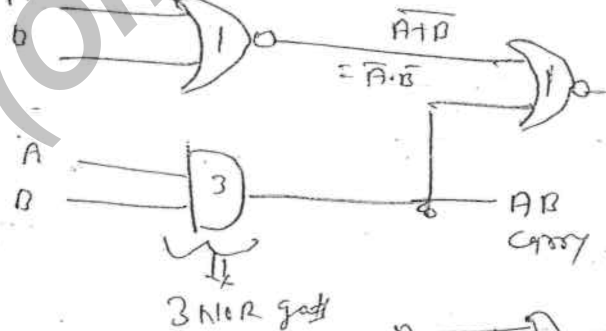


⇒ Half Adder using NOR gate

EXOR → $\bar{A}B + A\bar{B} = \overline{A \odot B}$

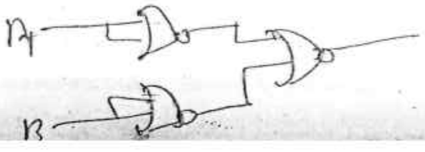
$= \overline{AB + \bar{A}\bar{B}}$

$=$

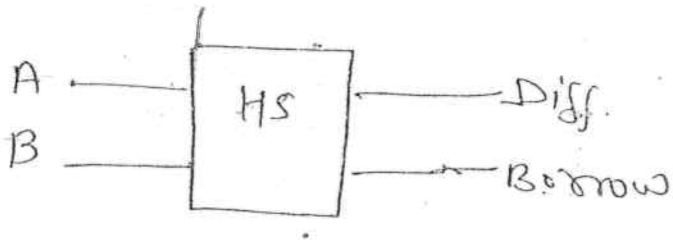


$\overline{AB + \bar{A}\bar{B}}$

$= A \oplus B$ sum



Half Subtractor

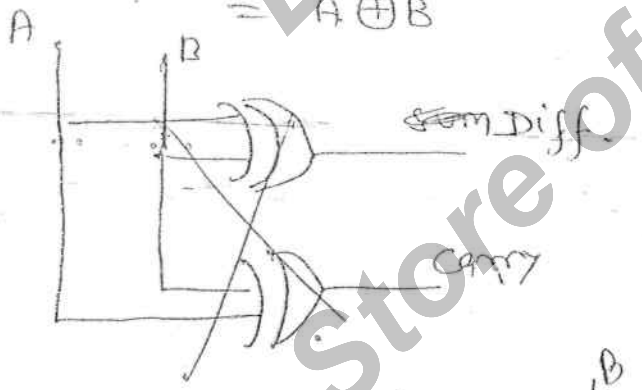


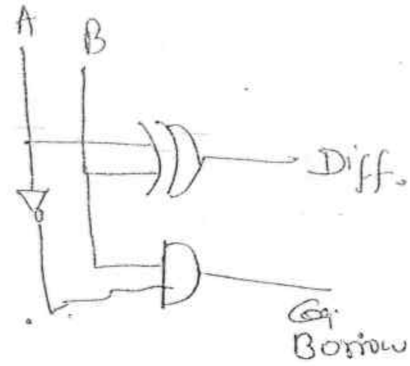
Truth table

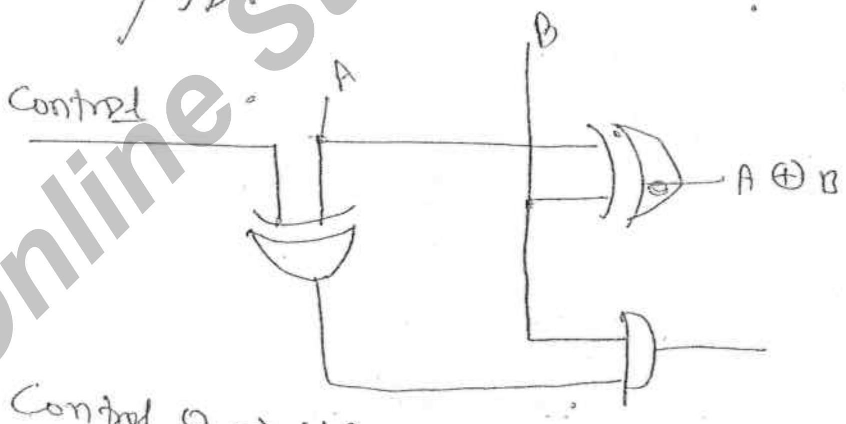
A	B	A − B Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$\text{Diff.} = \overline{A}B + A\overline{B}$ $\quad$ $\text{Borrow} = \overline{A}B$

$= A \oplus B$







Control 0 → HA
Control 1 → HS

HS

(1) Logical exp. for sum = $A \oplus B$

(2) " " carry = $\bar{A}B$

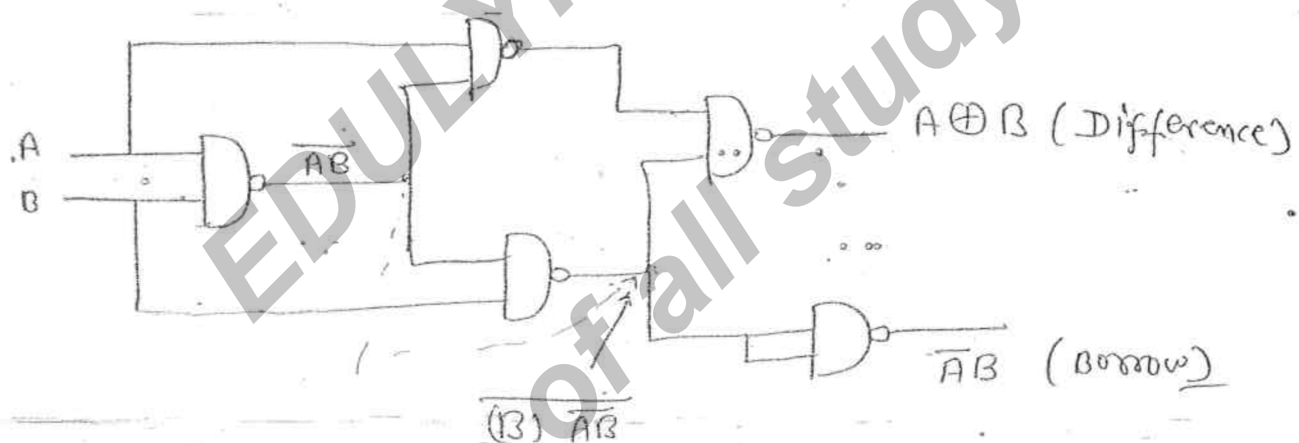
(3) min. No. of NAND = 5

(4) " " NOR = 5

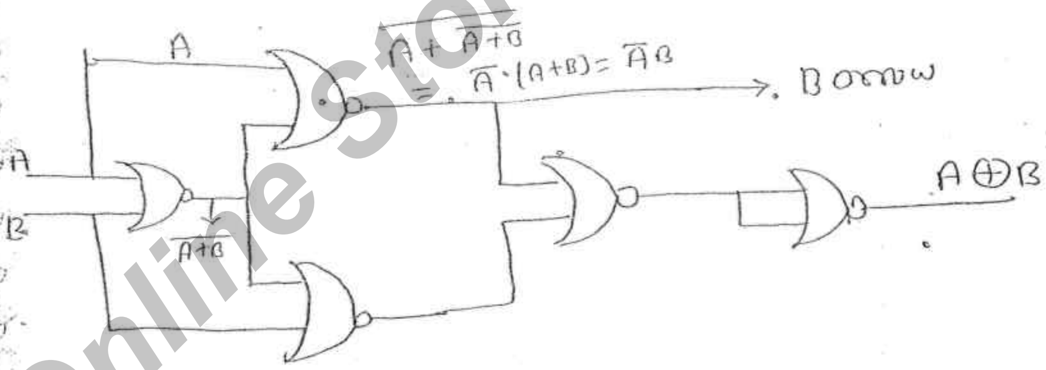
(5) No. of MUX —

(6) No. of Decoder —

HS using NAND :⇒

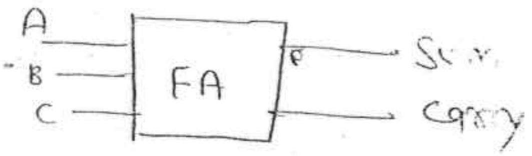


HS using NOR :-



Full Adder ckt

A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Logical expression

$SUM = A \oplus B \oplus C$

$= \sum m(1,2,4,7)$

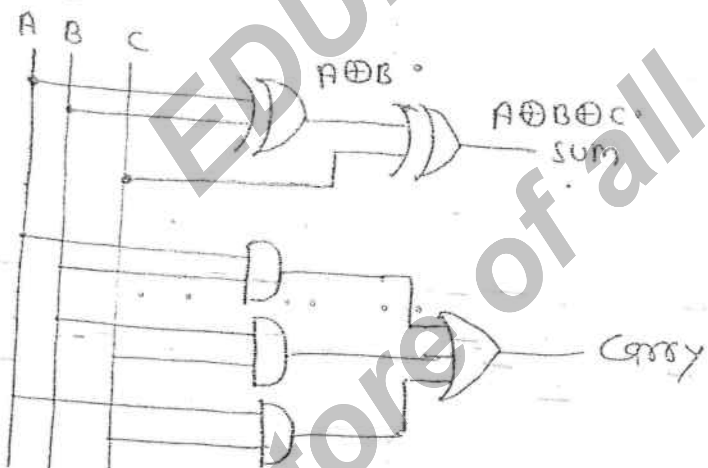
$Carry = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$

$Carry = \sum m(3,5,6,7)$

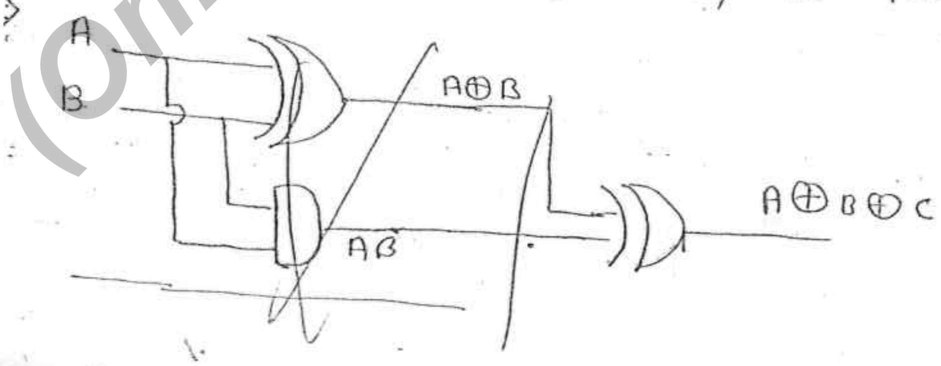
$\boxed{Carry = AB + BC + AC}$

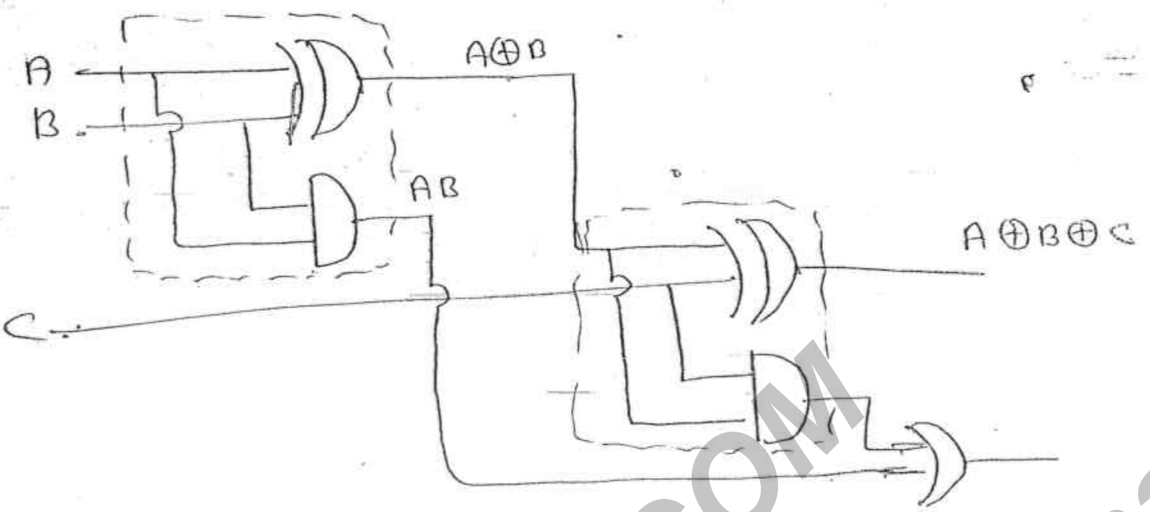
$= AB(C+\bar{C}) + (\bar{A}B + A\bar{B})C$

$= AB + (A \oplus B)C$



⇒ in full Adder if all logic gates have same propogation Delay than to provide sum output or carry output minimum $2t_{pd}$ delay is Required.





→ full adder can be implemented by using 2 half adder and one OR gate. (But delay is 3 tpd)

FA

(1) Sum = $\Sigma m(1,2,4,7)$ → $A \oplus B \oplus C$

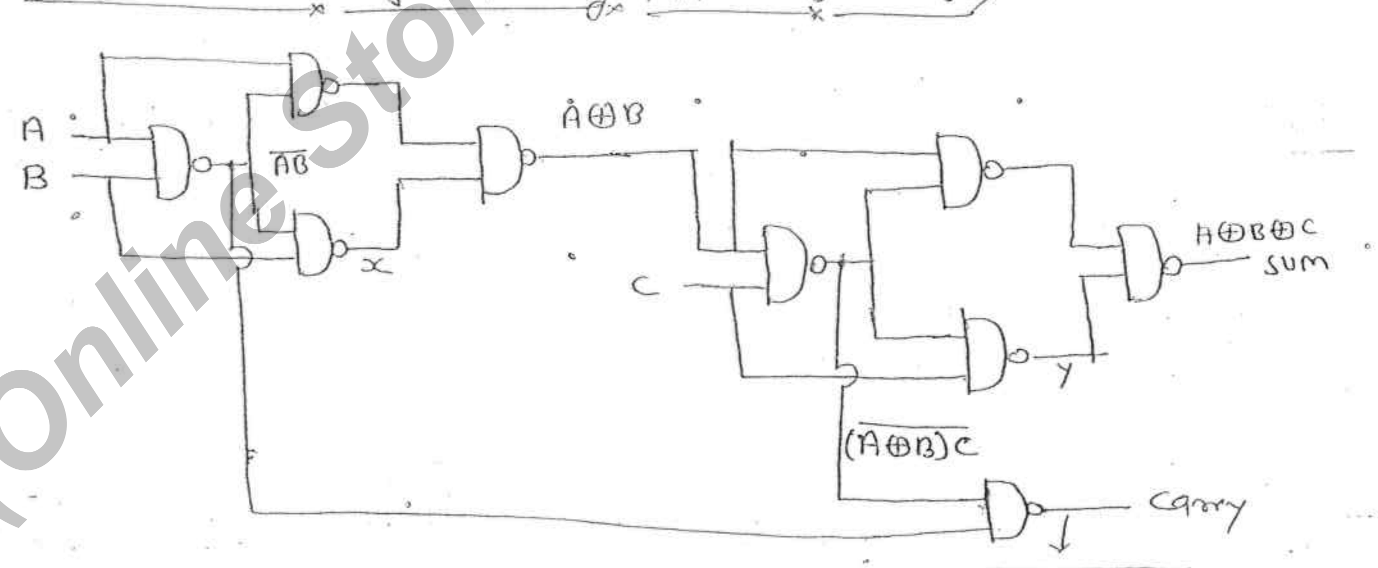
(2) Carry = $AB + BC + AC$ → $AB + (A \oplus B)C$ → $\Sigma m(3,5,6,7)$

(3) No. of HA and OR = 2 HA & 1 OR gate

(4) No. of NAND gate = 9

(5) No. of NOR gate = 9

⇒ Implementation of FA using NAND gate :-

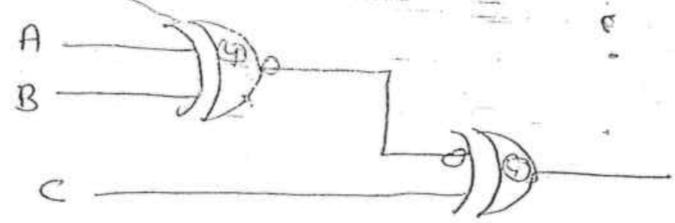


$\overline{\overline{AB} \cdot \overline{(A \oplus B)C}}$

$x = \overline{\overline{AB} \cdot B}$ $\quad y = \overline{\overline{(A \oplus B)C} \cdot C}$ $\quad = AB + (A \oplus B)C$

FA using NOR gate :⇒

$A\oplus B\oplus C$ =

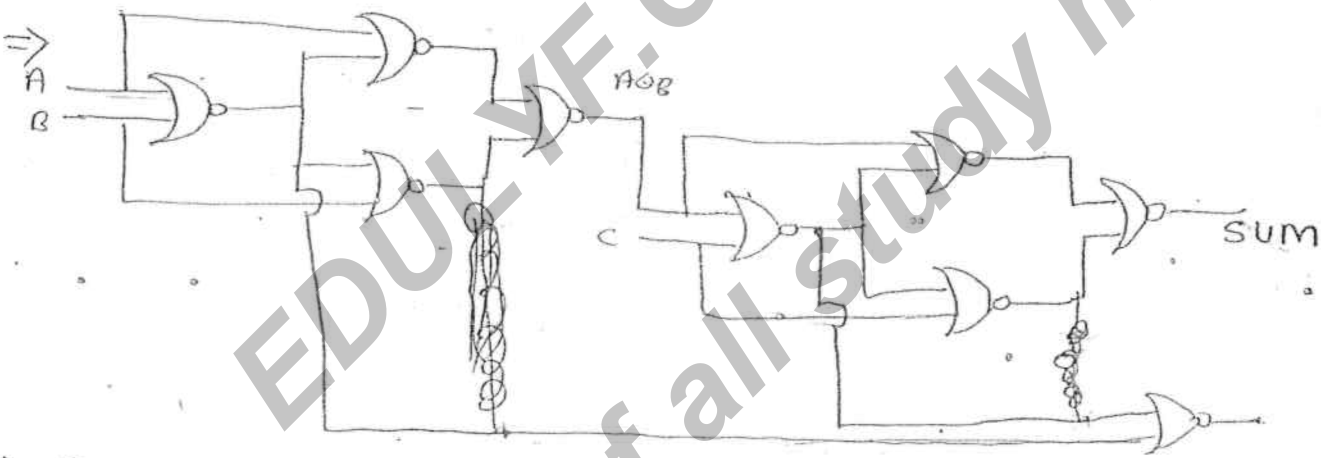


= $A\odot B\odot C$

$A\oplus B = \overline{A\odot B}$

$A\oplus B\oplus C = (A\odot B)\odot C$

⇒



⇒ Full Substractor :⇒

(i) Diff = $A\oplus B\oplus C$ $\Sigma m(1,2,4,7)$

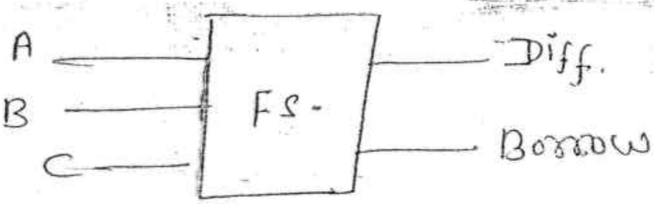
2) Borrow → $\bar{A}B + BC + \bar{A}C$

→ $\cancel{\bar{A}B + (\overline{A\oplus B})C}$ $\bar{A}B + (A\odot B)C$ $\Sigma(1,2,3,7)$

3) No. of HS & OR

4) No. of NAND gate → 9

5) No. of NOR → 9



A	B	C	$(A-B)-C$ Diff.	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Borrow $= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC$

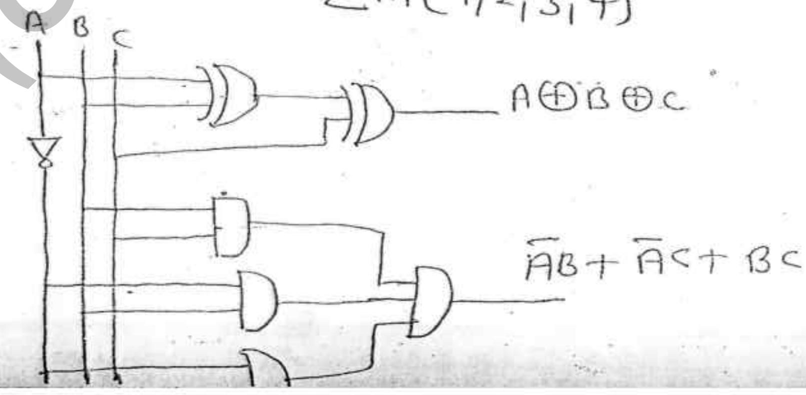
$= \bar{A}B + \bar{A}C + BC$ —— (1)

$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC$

$= \bar{A}B + (\bar{A}\bar{B} + AB)C$

$= \bar{A}B + (A \odot B)C$ —— (2)

$= \sum m(1,2,3,7)$



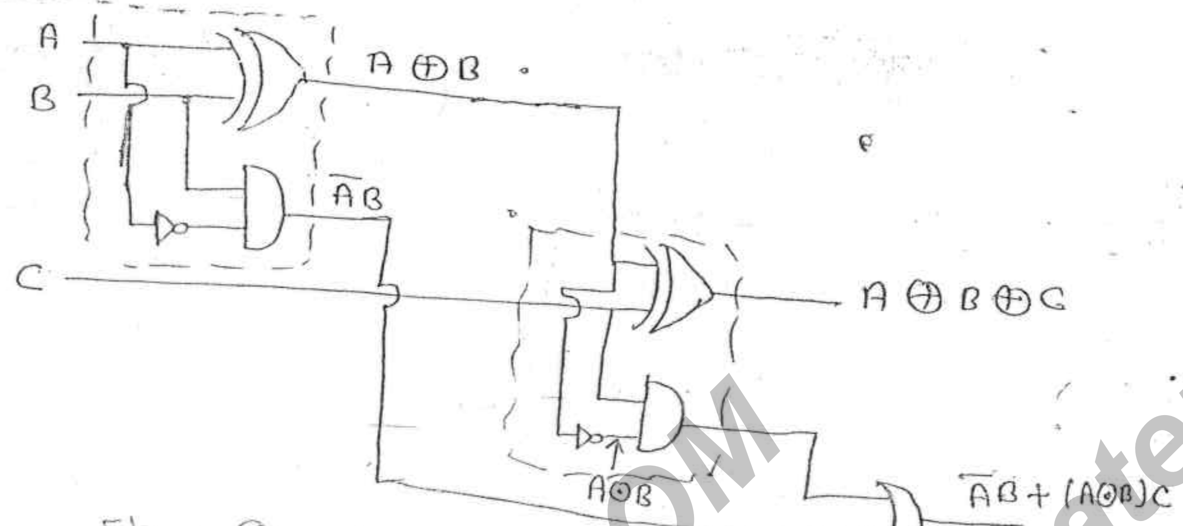


Fig. → FA using 2HA & 1 OR
FS 2HS

⇒

30/08/2011

To add group of bits following adder circuits are used (i) serial adder (ii) parallel adder (iii) Look ahed carry adder.

Serial adder

In serial adder to add n-bit numbers only one full adder is used....

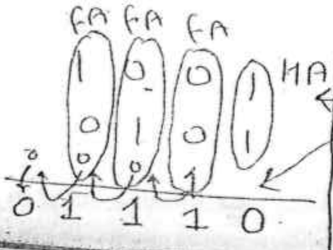
To provide n bit addition it requires minimum n clock pulses.

This is the slowest adder among all.

Parallel adder

→ in parallel adder to provide each bit addition seperate adder circuit is used.

In this to provide n bit addition it require (n-1)FA and 1 HA (OR) n FA (OR) (2n-1) HA and (n-1) OR gates.



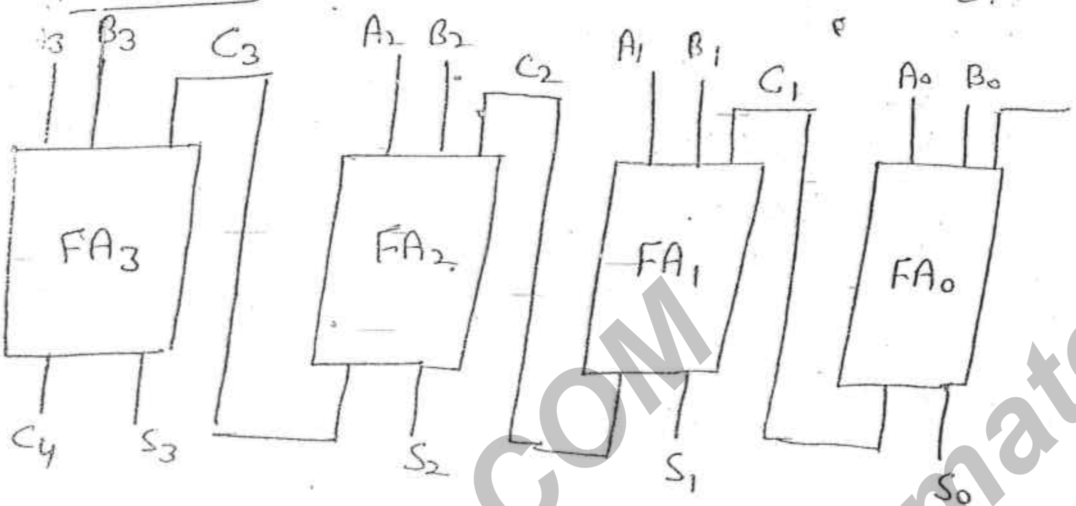
(i) 4-bit → 3FA & 1 HA

OR (ii) 4 FA if 0

(iii) ∵ 1 FA → 2 FA & 1 OR gat

(2n-1) &(n-1)

4 adder



$A_3\ A_2\ A_1\ A_0$

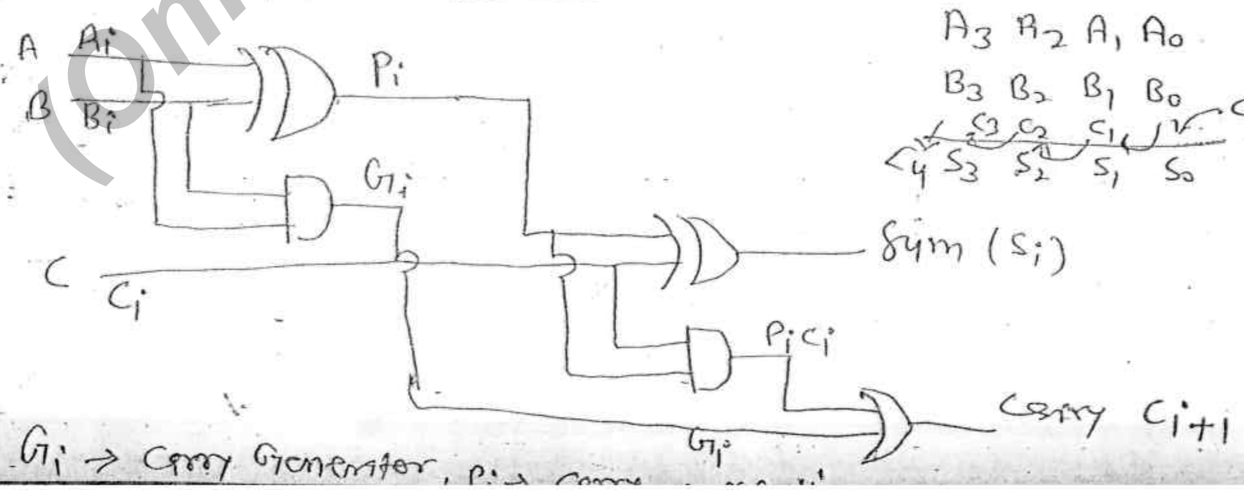
$B_3\ B_2\ B_1\ B_0 \leftarrow C_0$

$C_3\ C_2\ C_1$

$S_3\ S_2\ S_1\ S_0$

- In parallel adder each FA will provide two logic gate delays ($2t_{pd}$).
- To provide n-bit addition it require $2n\,t_{pd}$ delay
- in parallel adder carry propagation delay is present from input to output Hence it is also known as ripple carry adder.
- The Disadvantage of parallel adder is as No. of stage / bit is increases the speed of operation is decreases

Look Ahead Carry Adder :→



$A_3\ A_2\ A_1\ A_0$

$B_3\ B_2\ B_1\ B_0 \leftarrow C_0$

$C_3\ C_2\ C_1$

$C_4\ S_3\ S_2\ S_1\ S_0$

$G_i \rightarrow$ Carry Generator, $P_i \rightarrow$ Carry ...

$$P_i = A_i \oplus B_i$$
$$G_i = A_i B_i$$
$$S_i = P_i \oplus C_i$$
$$C_{i+1} = P_i C_i + G_i$$

$P_0 = A_0 \oplus B_0$	$G_0 = A_0 B_0$	$S_0 = P_0 \oplus C_0$
$P_1 = A_1 \oplus B_1$	$G_1 = A_1 B_1$	$S_1 = P_1 \oplus C_1$
$P_2 = A_2 \oplus B_2$	$G_2 = A_2 B_2$	$S_2 = P_2 \oplus C_2$
$P_3 = A_3 \oplus B_3$	$G_3 = A_3 B_3$	$S_3 = P_3 \oplus C_3$

$C_0 \rightarrow$ input carry

$\rightarrow C_1 = P_0 C_0 + G_0$

$C_2 = P_1 C_1 + G_1$

$C_2 = P_1 (P_0 C_0 + G_0) + G_1$

$\rightarrow C_2 = P_1 P_0 C_0 + P_1 G_0 + G_1$

$C_3 = P_2 C_2 + G_2$

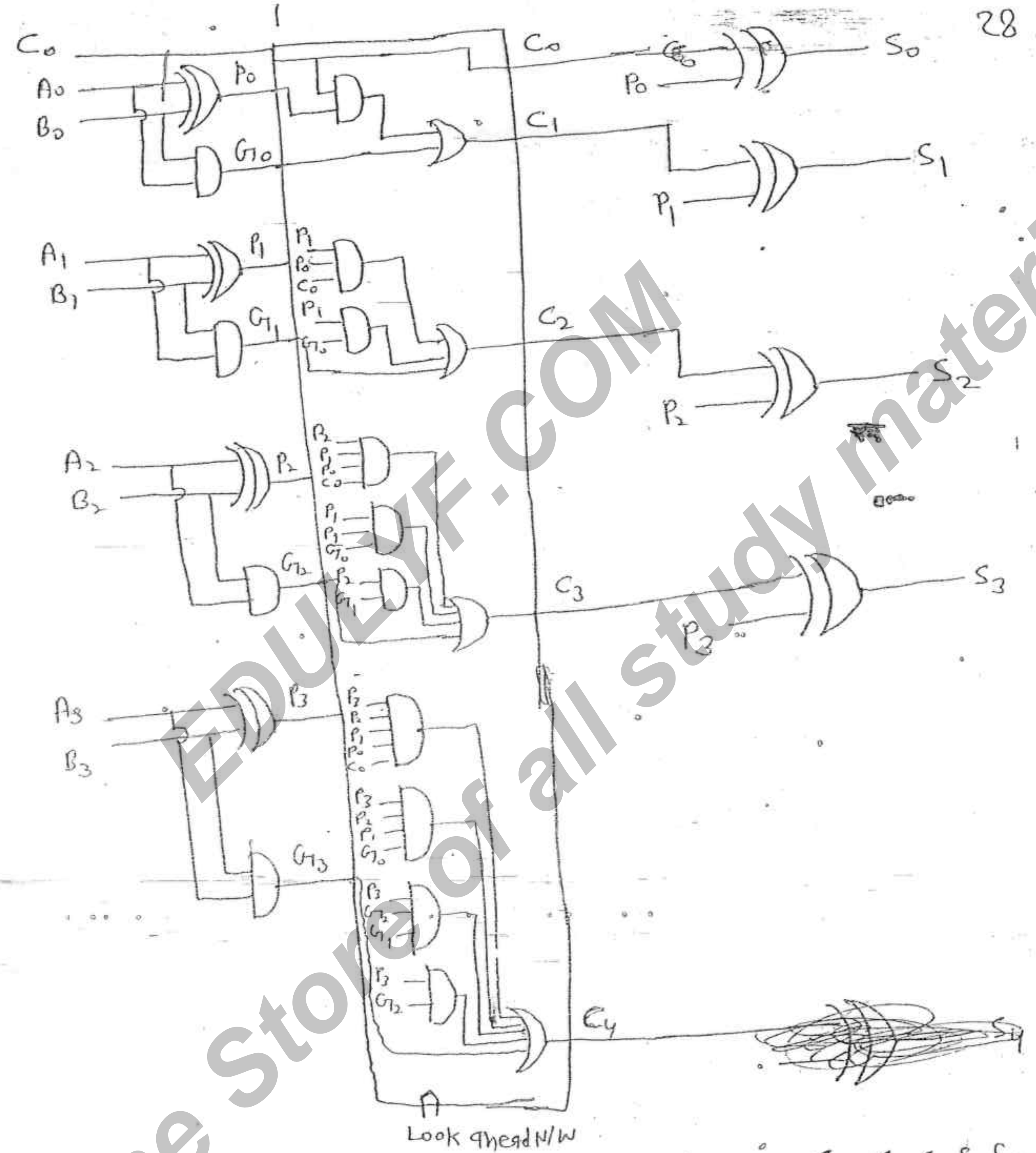
$\rightarrow C_3 = P_2 P_1 P_0 C_0 + P_2 P_1 G_0 + P_2 G_1 + G_2$

$C_4 = P_3 C_3 + G_3$

$+P_3 G_2$

$\rightarrow C_4 = P_3 P_2 P_1 P_0 C_0 + P_3 P_2 P_1 G_0 + P_3 G_2 G_1 + G_3$

Look ahead Carry N/W



sum ⇒ $C_4 S_3 S_2 S_1 S_0$.

Fig. ⇒ Look Ahead carry adder.

→ Look Ahead carry Adder is fastest adder in this carry is generated in parallel by Look ahead carry Network.

→ In this to provide carry output 3 t_{pd} delay and to provide sum output delay 4 t_{pd} delay is required.

→ Look ahead carry N/W contains two level AND-OR gates.

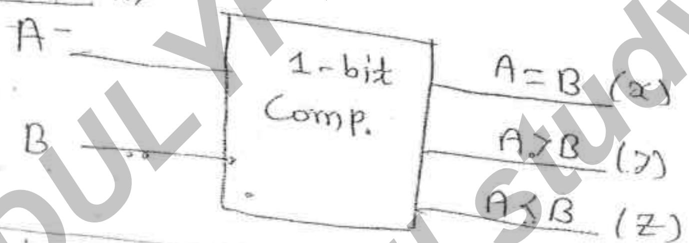
→ No. of AND gates used in Look Ahead carry adder is $= \frac{n(n+1)}{2}$.

→ No. of OR gates used $= n$

→ Total No. of AND gates $= \frac{n(n+1)}{2} + n$

Comparator

1-bit Comparator :→

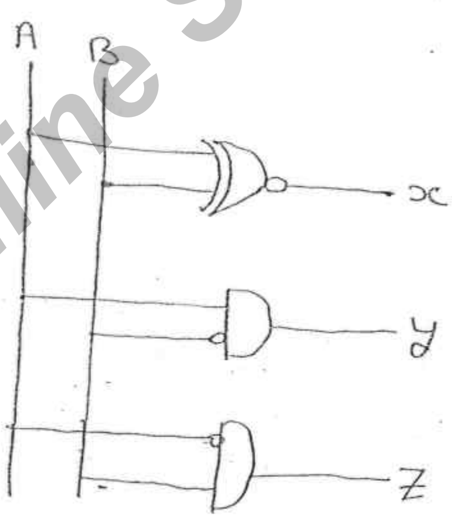


A	B	x	y	z
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

$x = \bar{A}\bar{B} + AB = A \odot B$ ⇒ (A=B)

$y = A\bar{B}$ ⇒ (A>B)

$z = \bar{A}B$ = (A<B)



two i/p → three o/p